Deterministic Three-Tensor Isomorphism over Finite Fields

1 October 2026

Abstract

For a finite field of odd characteristic, we give a deterministic algorithm for isomorphism of arbitrary three-tensors of format at most $n \times n \times n$. Its running time is $q^{O(n \log^3(n+2))} \operatorname{poly}(\log q)$, including an actual isomorphism and generators of the full source stabilizer. The argument retains the original faithful group degree while moving zero-block interiors one physical factor at a time. Natural-degree algebra ideals, covariant coefficient value flags, and exact normal-p-by-unitary transport keep all coefficient lifts and tensor shears. Three physical passes decrease the largest active boundary simple degree. A separate fixed-round variant sharpens the bound to $q^{O(n \log(n+2))} \operatorname{poly}(\log q)$. This note does not claim a $q^{O(n)}$ bound.

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1 The theorem and its conventions

Let $k = \mathbb{F}_q$ have odd characteristic. A three-tensor is a trilinear form $T : V_1 \times V_2 \times V_3 \to k$. The factor order is fixed. An isomorphism from T to T' is a triple $g = (g_1, g_2, g_3)$ of invertible factor maps such that

$$T'(x_1, x_2, x_3) = T(g_1^{-1}x_1, g_2^{-1}x_2, g_3^{-1}x_3).$$

We assume the two formats agree; otherwise the answer is negative. Empty factors and the all-zero format are immediate and may be removed from the discussion.

Theorem 1.1. *For arbitrary three-tensors over an explicitly represented finite field $k = \mathbb{F}_q$ of odd characteristic, put $n = \max_i \dim V_i$. There is a deterministic algorithm of running time*

$$q^{O(n \log^3(n+2))} \operatorname{poly}(\log q)$$

which decides isomorphism. If the transporter is nonempty, it returns an actual factor-map triple and generators for the complete source stabilizer in $\prod_i \mathrm{GL}(V_i)$. The bound is worst-case, without a rank, nonsingularity, conciseness, or randomness promise.

The polynomial in the tensor input length is absorbed into the displayed exponential term. We prove a bound $q^{O(N\log^3(N+2))}\,\mathrm{poly}(L,\log q)$, where $N=O(n)$ is the original faithful group degree and L is the explicit auxiliary input length. Every auxiliary module used below has size bounded by a fixed polynomial in the original input length. In particular, a large multiplicity space is never substituted for N.

A transporter is returned as one representative and generators of its source stabilizer. Words in earlier generators are stored as straight-line programs. When several complete transporter cosets cover the answer, one successful representative, all their source generators, and ratios of their representatives recover the full transporter. Every ratio used in this construction satisfies the full original equation.

1.1 A single scalar anchor

Write $A^{\mathrm{ex}}=A\times A^{\mathrm{op}}$ for the exchange algebra with $(a,b)^*=(b,a)$. Its unitary group is $\{(g,g^{-1}) : g\in A^\times\}$. Initialize the algebra as

$$E_{\mathrm{init}}=\prod_{i=1}^{3}\mathrm{End}_k(V_i)^{\mathrm{ex}}\ \times k^{\mathrm{ex}}.$$

It is faithfully represented on the physical spaces, their paired duals, and two scalar coordinates. Let $\chi : E_{\mathrm{init}}\to k$ be the primal coordinate of the last factor. Its restrictions to every subsequent subalgebra are unital linear algebra homomorphisms. The initial unitary group is the full physical group times $k^\times$.

For the proof, this extended group acts on full tensors by

$$(g\star T)(x_1,x_2,x_3)=\chi(g)T(g_1^{-1}x_1,g_2^{-1}x_2,g_3^{-1}x_3). \tag{1}$$

At the end we intersect the complete transporter with $\chi=1$. This is an explicit character-kernel calculation, so it returns precisely the original tensor transporter. The anchor is reused throughout; it is not replicated at successive marks.

The algorithm retains physical subspaces $T_i\le V_i$ such that

$$T(T_1,T_2,T_3)=0. \tag{2}$$

We may initialize $T_1=0$, $T_2=V_2$, and $T_3=V_3$; all three zero also works. Thus the state exists for every input without guessing a zero block. Put $C_i=V_i/T_i$. The full group in the state preserves all current and stored physical flags. On quotient coefficient vectors use

$$\rho_i(g)c=\chi(g)^{-1}g_ic. \tag{3}$$

This scalar normalization does not alter the action on subspaces.

For $c\in C_i$, define $M_c^T=T(c,-,-)$ on the other two interiors. It is independent of the lift of c by (2). Let $R_i(g)$ be ordinary inverse pullback on its two row arguments. Direct substitution gives

$$M^{g\star T}_{\rho_i(g)c}=R_i(g)M_c^T. \tag{4}$$

Consequently the pair (M,c) has the exact action $(M,c)\mapsto(R_i(g)M,\rho_i(g)c)$. This identity is used at every quotient mark. It prevents a coefficient lift from being chosen independently of its contraction shift.

1.2 The two parameters

The represented algebra and actual group always use degree $N = O(n)$. Choose a sufficiently large absolute matrix-size cutoff a_0. If a simple factor of the current pure algebra on a boundary module is $\mathrm{Mat}_a(\mathbb{F}_{q^f})$, its natural degree is $\delta = fa$. It is called *active* when $a > a_0$. The round parameter is the largest active boundary degree. A commutative extension field can have large δ but is not active. Under algebra restriction, a child matrix size cannot exceed its parent size, so an inactive type cannot become active.

The proof uses physical passes: during a pass in direction i, only T_i may grow, while the other two interiors shrink. This ordering is what makes the actual partial ideal-image potentials monotone. Three passes reduce the largest active boundary degree by a fixed factor. The final supported-tensor algorithm applies once all boundary matrix sizes are bounded.

2 Exact algebra and group primitives

All algebras are finite-dimensional unital k-algebras. An involution is k-linear and reverses multiplication. We write

$$\mathrm{U}(E) = \{u \in E : u^*u = uu^* = 1\}.$$

A genuine bimodule has commuting unital left and right algebra actions; its unitary action is $v \mapsto uvu^{-1}$. An invariant subquotient is supplied by actual numerator and denominator spaces and the induced linear maps. Invariance under the group does not mean invariance under both algebra actions.

Proposition 2.1 (Represented algebra operations)**.** *For a star-algebra faithfully represented in degree N, radical and semisimple structure, an actual star-stable Levi complement, deterministic module isomorphism, unitary norm correction, and generators of the full unitary group can be computed in $q^{O(N)}\,\mathrm{poly}(L, \log q)$ time. The same bound applies to its represented subalgebras. Modules of polynomial auxiliary dimension enter only the polynomial factor.*

Proof. We use the deterministic finite-algebra and star-algebra algorithms in [5, 1, 3], together with module isomorphism [2, Corollary 3 of the authors' manuscript]. The permitted polynomial dependence on the characteristic is absorbed in $q^{O(N)}$; see [1, Section 7.4] and [3, Theorem 7(1), arXiv version].

For clarity, the representation degree used by structure calculations is the original one. A composition series of its natural module can be found by enumerating vectors and computing cyclic spans. A proper submodule contains a vector with a proper cyclic span, so at most N searches in spaces of dimension at most N suffice. The radical is the kernel of the simultaneous action on all simple sections. Every simple algebra type occurs because the representation is faithful. Its field degree and matrix size are recovered from the simple representation and its commutant. Standard separability and matrix-unit lifting construct a Levi complement [4]; the odd-characteristic star correction gives a star-stable one. Thus these steps do not enumerate a space of size $q^{\dim E}$.

For $J = \mathrm{rad}\, E$ and a star-Levi F,

$$\mathrm{U}(E) = \mathrm{U}(1 + J) \rtimes \mathrm{U}(F). \tag{5}$$

The unitary radical is a p-group. Cayley parametrization

$$a \longmapsto (1 - a)(1 + a)^{-1}, \qquad a^* = -a, \quad a \in J,$$

and the radical filtration give actual layer lifts and generators. Semisimple norm correction reduces to the usual finite-field classical form problems; nilpotent corrections are obtained by the odd-characteristic square-root procedure. These are the precise norm and generation operations used below. □

Lemma 2.2 (Fixed vectors and subspace conductors)**.** *Pure unitary transport of a fixed vector in a genuine bimodule, or of a finite family of subspaces in genuine one-sided modules, can be computed with full stabilizers in $q^{O(N)}\operatorname{poly}(L, \log q)$ time. A nonempty answer is a coset of the unitary group of an explicitly computed star-subalgebra of E.*

Proof. For the subspace version, first compare dimensions and set

$$F = \{a : aL_j \subseteq L_j,\ a^*L_j \subseteq L_j\ \forall j\}, \qquad X = \{c : cL_j \subseteq L'_j,\ c^*L'_j \subseteq L_j\ \forall j\}.$$

Then F is a unital star-subalgebra and X is a right F-module. If a unit $c_0 \in X$ exists, equal dimensions turn all inclusions into equalities and $X = c_0F$. Test $F_F \cong X_F$ by deterministic module isomorphism. For a module generator c, testing invertibility in the original representation is conclusive: if a unit generator existed, every generator would differ from it by an F-unit. For a unit c, the element c^*c belongs to F. Norm correction in F decides whether cF contains a unitary; its complete unitary transporter is then $u\,\mathrm{U}(F)$.

For a bimodule vector v and target w, use instead

$$F = \{a : av = va,\ a^*v = va^*\}, \qquad X = \{c : cv = wc,\ c^*w = vc^*\}.$$

The same right-module argument and norm correction apply. These spaces have dimension at most $\dim E \le N^2$; large supplied modules contribute only linear equations. This argument does not use an invertible-element oracle for an arbitrary matrix space. □

Lemma 2.3 (Original-degree group operations)**.** *Membership, generator reduction, pointwise kernels of explicit representations of degree $O(N)$, normal closure, and elementary-abelian image/kernel computations for the actual generated groups used here cost $q^{O(N)}\operatorname{poly}(L, \log q)$.*

Proof. Use their faithful action on the original vector space as a permutation representation of degree q^N, and deterministic permutation-group routines [6]. A bounded direct sum of original-degree representations still has degree $O(N)$. This permits homomorphism kernels by fixing the target permutation domain of degree $q^{O(N)}$ pointwise, without enumerating the quotient group. Arbitrary polynomial auxiliary representation degree is not permitted in this argument; auxiliary additive kernels use the following explicit construction. Images enumerated elsewhere are only those with an independently proved order $q^{O(N)}$: boundary, field, factor-permutation, and scalar-character images. Normal intersections and preimages stay in the original permutation action, as in [10, Section 3].

For an explicit homomorphism $\eta : G \to A_+$, compute the prime-field span of generator images and actual words lifting a basis. The entire kernel is the normal closure of corrected original generators, commutators of basis lifts, and their pth powers. The quotient by these relators is exactly the specified elementary abelian image. All words remain straight-line programs. □

The uniform base-field rank-cover inequality and its deterministic canonical construction are proved in Lemma 3.7 and Proposition 3.8. We use the factor 4, without a field-size assumption; a generic-rank factor-2 statement is not substituted for this base-field hypothesis.

The rank-cover spaces below are physical row subspaces of total dimension $O(N)$. Their actual-group transport is supplied by Corollary E.4, which adds only the two current physical frame groups. Older flags remain inside the actual group and are never re-encoded by one free frame per level.

3 Natural-degree selection and covariant value flags

All logarithms in this section have base two. Let $E \subseteq \mathrm{End}_{\mathbb{F}_q}(W)$ be a faithful unital algebra, where $\dim W = N$. Write

$$E/J(E) = \prod_i A_i, \qquad A_i = \mathrm{Mat}_{a_i}(\mathbb{F}_{q^{f_i}}), \qquad D_i = f_i a_i^2, \quad \delta_i = f_i a_i. \tag{6}$$

Let d_i be the multiplicity of the natural simple A_i-module in W. Faithfulness implies $d_i \geq 1$, and $\sum_i d_i \delta_i = N$. We use the standard Wedderburn–Artin and Wedderburn–Malcev structure theorems and separability of finite-field algebras; see [4]. The cutoff below concerns δ_i, not D_i.

3.1 Cutoff ideals and an entropy potential

For an integer $\tau \geq 1$, let

$$\mathcal{L}_\tau(E) = \{i : \delta_i > \tau\}.$$

Lift the central idempotent supported on these factors to an idempotent $e \in E$, and put $I_\tau(E) = EeE$. For every genuine E-module V, put $A_\tau(E, V) = I_\tau(E)V$.

Lemma 3.1. *The ideal $I_\tau(E)$ is independent of the lift. If $E' \subseteq E$ is a unital subalgebra, then $I_\tau(E') \subseteq I_\tau(E)$. Every simple factor of $E/I_\tau(E)$ has natural degree at most τ. These assertions hold on every supplied module, without a faithfulness assumption on that module.*

Proof. If e, e' lift the same quotient idempotent, then $c = e'e + (1 - e')(1 - e)$ belongs to $1 + J(E)$ and satisfies $ce = e'c$. Thus the generated ideals agree. A simple algebra B with natural degree $\delta(B)$ acting nontrivially on an A_i-natural module has $\delta(B) \leq \delta_i$: its identity cuts out a nonzero module which is a sum of natural B-modules. Choose a semisimple complement of E'. A factor B of that complement with $\delta(B) > \tau$ cannot inject into the semisimple quotient of $E/I_\tau(E)$, since each of its factors has natural degree at most τ. Its image in $E/I_\tau(E)$ is therefore zero. Its identity, and then the ideal generated by all such identities, belong to $I_\tau(E)$. The description of the remaining simple factors follows directly from the defining quotient idempotent. □

Define

$$\Phi(E, W) = \sum_i d_i \delta_i \log D_i, \qquad \Psi_\tau(E, W) = \Phi(E, W) + \tau |\mathcal{L}_\tau(E)|. \tag{7}$$

Then $0 \leq \Psi_\tau(E, W) \leq 2N \log N + N$ (with the evident convention when $N = 1$). Indeed, $D_i \leq N^2$, and each counted factor uses more than τ dimensions of the faithful module.

Proposition 3.2. *Both potentials in* (7) *are nonincreasing under unital subalgebra restriction. Form the bipartite graph of old and new simple factors with natural degree greater than τ, joining two factors when the new factor occurs on an old natural module. Every new vertex has a neighbor. If*

$$s = \sum_{i \ old} (\deg i - 1)_+, \quad t = \sum_{j \ new} (\deg j - 1), \quad z = |\{i \ old : \deg i = 0\}|,$$

then

$$\Phi(E, W) - \Phi(E', W) \geq 2\tau s, \qquad \Psi_\tau(E, W) - \Psi_\tau(E', W) \geq \tau(s + t + z).$$

Proof. Let L' be a semisimple complement of E'. Its intersection with $J(E)$ is zero. Each simple factor of L' therefore has an injective projection to some old factor, and the preceding natural-degree observation proves the neighbor assertion.

On one natural module for $A = \mathrm{Mat}_a(K)$, $K = \mathbb{F}_{q^f}$, write t_B for the K-rank of the identity of a factor B of L'. These identities are orthogonal, so

$$\sum_B t_B = a, \qquad \dim_{\mathbb{F}_q} e_B K^a = f t_B, \qquad \dim_{\mathbb{F}_q} B \le f t_B^2. \tag{8}$$

The contribution of this old natural module to the entropy loss is

$$\sum_B f t_B \log \frac{f a^2}{\dim B}. \tag{9}$$

Every term is nonnegative. If there are $h \ge 2$ large new neighbors, all but one have $t_B \le a/2$. Each of these $h-1$ terms is at least $2ft_B > 2\tau$, since its corner contains a natural B-module of degree greater than τ. Faithfulness supplies an old natural-module copy for every old vertex, proving the first graph inequality. Edge counting gives $K_{\mathrm{new}} - K_{\mathrm{old}} = s - t - z$, where K counts large factors. Adding the τK term proves the second inequality and monotonicity. □

All comparisons are exact integer comparisons: one may store

$$2^{\tau |\mathcal{L}_\tau(E)|} \prod_i D_i^{d_i \delta_i},$$

whose binary length is $O(N \log(N+2))$.

3.2 Near-full corners and the intrinsic relative lemma

Lemma 3.3. *If* $\mathrm{Mat}_b(\mathbb{F}_{q^g})$ *embeds, possibly nonunitally, in* $\mathrm{Mat}_a(\mathbb{F}_{q^f})$ *and* $h = \gcd(f, g)$*, then*

$$a \ge bg/h, \qquad \frac{gb^2}{fa^2} \le \frac{h^2}{fg}.$$

If $gb^2 > fa^2/2$*, then* $g = f$*,* $b > a/\sqrt{2}$*, and the identity of the smaller algebra has rank exactly* b *over* $\mathbb{F}_{q^f}$*. On that corner the smaller algebra is the full endomorphism algebra.*

Proof. Extend scalars to $K = \mathbb{F}_{q^f}$. The algebra $K \otimes_{\mathbb{F}_q} \mathrm{Mat}_b(\mathbb{F}_{q^g})$ is a product of h copies of $\mathrm{Mat}_b(\mathbb{F}_{q^{\mathrm{lcm}(f,g)}})$. Each natural module has K-degree bg/h, proving the inequalities. Writing $f = hu, g = hv$ with $\gcd(u, v) = 1$, distinct centers give $uv \ge 2$ and hence the ratio at most $1/2$. Otherwise $g = f$ and $b > a/\sqrt{2} > a/2$. The nonzero corner degree is a multiple of b and at most a, so it is exactly b; equality of dimensions gives the last assertion. □

The row space used in a pass need not faithfully represent E. We state the lemma in the form that is actually used. Let V be a genuine subquotient of W whose composition multiplicities d_i^V are at most d_i. In the tensor application

$$V = T_j \oplus T_k \oplus T_j^* \oplus T_k^*.$$

The partial forms are represented intrinsically on this module, not zero-extended through arbitrary complements of physical subspaces.

Theorem 3.4. *Let $z \mapsto M_z \in \mathrm{End}_{\mathbb{F}_q}(V)$ be a linear pencil on a coefficient space Z. For each z let $E_z \subseteq E$ be a supplied unital subalgebra whose action on V centralizes M_z. Set $\tau = 10r$, where $r \geq 1$ is an integer. Suppose, for every z, that*

$$\Psi_\tau(E, W) - \Psi_\tau(E_z, W) < r, \tag{10}$$

$$\dim I_\tau(E)V - \dim I_\tau(E_z)V < r. \tag{11}$$

There is a successive flag of linear coefficient kernels, with total codimension at most

$$C_V = \sum_{i \in \mathcal{L}_\tau(E)} f_i (d_i^V)^2,$$

whose final term Z_0 satisfies

$$\mathrm{rank}(M_z|_{I_\tau(E)V}) < r \qquad (z \in Z_0).$$

The value maps and kernels are computable deterministically in $q^{O(N+\dim Z)}\,\mathrm{poly}(N, \log q)$ time. No equality $E_z = E \cap \mathrm{Cent}(M_z)$ is required.

Proof. Put $A = I_\tau(E)V$ and $A_z = I_\tau(E_z)V$. By Proposition 3.2, the large-factor graph is a perfect matching, and $\Delta\Phi_z < r$. Denote the new partner of A_i by B_i. If $\dim B_i \leq D_i/2$, its own corner term in (9) exceeds τ: the corner has degree at least $\delta(B_i) > \tau$. This contradicts $\Delta\Phi_z < r$. Thus Lemma 3.3 gives

$$B_i = \mathrm{Mat}_{m_i}(\mathbb{F}_{q^{f_i}}), \qquad m_i > a_i/\sqrt{2}.$$

The projection of B_i to any other old large factor is zero by matching, and to an old small-natural-degree factor is zero by the natural-degree inequality. Every other child corner in A_i has rank at most $a_i - m_i$, so its algebra dimension is less than $f_i m_i^2$. Consequently

$$k_z := \sum_{i \in \mathcal{L}_\tau(E)} f_i d_i^V (a_i - m_i) \leq \tfrac{1}{2}\Delta\Phi_z < r/2. \tag{12}$$

Here the entropy is computed on W; its loss dominates the corresponding loss on V, since $d_i^V \leq d_i$ and every restriction summand is nonnegative. Write $\ell_z = \dim A - \dim A_z < r$.

Fix a semisimple complement L of E and choose its large-factor idempotent e. A semisimple subalgebra $B \subseteq E$, including one with a smaller identity, can be conjugated into L by $u \in 1 + J(E)$, keeping its image modulo $J(E)$. For clarity, this relative form follows directly from separability. If $\sigma, \eta : C \to E$ are two lifts of a unital semisimple algebra with the same quotient image, and $\sum x_j y_j = 1$, $\sum cx_j \otimes y_j = \sum x_j \otimes y_j c$ is a separability element, then

$$u = \sum_j \sigma(x_j)\eta(y_j) \equiv 1 \pmod{J(E)}, \qquad \sigma(c)u = u\eta(c).$$

Adjoining the complementary scalar identity treats the nonunital case.

Apply this to the product of the matched factors B_i of a complement of E_z. On V write

$$E' = u^{-1}E_z u, \quad M' = u^{-1}M_z u, \quad A' = u^{-1}A_z,$$

and let e_B be the identity of the resulting product $B \subseteq L$. Then $A' = E'e_B V$, $e_B V \subseteq A' \subseteq A$, and $\dim A/A' = \ell_z$. Since M' commutes with B, its restriction to $e_B V$ is a $d_i^V \times d_i^V$ matrix over $\mathbb{F}_{q^{f_i}}$ on

the i-part. Extend its scalar blocks from m_i to a_i natural coordinates. This gives $w \in \mathrm{End}_L(V)$ supported on the old large types, with

$$M'e_B = we_B = e_B w. \tag{13}$$

The space of possible w has dimension C_V.

Choose a module projection $p : A \to A'$ for the semisimple algebra $B + \mathbb{F}_q(1 - e_B)$, and put $G = M'p$. The image identities are

$$\operatorname{im} G = M'A' = E'M'e_B V = E'we_B V.$$

Set $Y = (1 - e)G + w|_A$. Since $e_B p = pe_B = e_B$ on A, the image of $eG - w|_A$ lies in $(e - e_B)V$, of dimension k_z. Thus

$$M'|_A = Y + R, \qquad \operatorname{rank} R \le \ell_z + k_z < 2r, \qquad eY = w|_A. \tag{14}$$

Both G and Y have image in A.

Choose an E-stable flag containing A and having semisimple successive quotients, and split it using L. The matrix u is upper block unitriangular. If the output rows of w after a flag term V_a vanish, then $\operatorname{im} w \subseteq V_a$; the displayed image identity for G implies $\operatorname{im} G \subseteq V_a$, and hence $\operatorname{im} Y \subseteq V_a$. Process large output blocks in reverse flag order and input blocks in forward flag order. On the current kernel all later output values and all earlier input values in the current row are zero. Therefore the current large-type block of uYu^{-1} is exactly the corresponding block of w. The other terms in that block multiplication vanish by the image property and $eY = w|_A$.

In individual natural-copy coordinates that block is multiplication by a field element, within error rank less than $2r$. A nonzero difference of two such multiplication maps has rank $\delta_i > 10r$. The coefficient is therefore unique. Comparing three matrices proves additivity, since any failure would have rank less than $6r$; scalar homogeneity is proved the same way. Pass to the kernel of this actual linear coefficient map and continue. Uniqueness holds for every allowed normalization, so the induction does not depend on a choice of u or p. There are at most C_V field coordinates. On the final kernel $w = 0$, whence $M'A' = 0$ by the image identity. Thus $M_z A_z = 0$, and $\operatorname{rank}(M_z|_A) \le \ell_z < r$.

For computation, module composition data are obtained by cyclic-vector enumeration in modules of degree at most N, and linear algebra: a nonzero vector with smallest positive cyclic span generates a simple submodule. Repeat on the quotient. Intersecting the kernels of the simple representations gives $J(E)$; equal kernels identify repeated simple types. The dimensions of their image algebras give D_i, and $f_i = \delta_i^2/D_i$, $a_i = D_i/\delta_i$. Chinese remaindering in these quotient representations gives the large central idempotent image. If x lifts that image, apply the polynomial which is zero modulo t^N and one modulo $(t-1)^N$ to obtain an idempotent lift.

On each semisimple flag quotient, linear intertwiner equations give identifications of repeated simple copies and their finite division endomorphism fields. A projection onto a known summand is found by linear equations, so these semisimple modules can be split without enumerating a large matrix algebra. These are also standard applications of module-isomorphism algorithms [2]. Choose arbitrary lifts of these quotient coordinates to V. An L-split choice differs by an upper block-unitriangular coordinate change. The preceding block cancellation therefore proves that the extracted values are the same; an actual semisimple complement need not be constructed for extraction.

For an explicit coefficient calculation, enumerate its center field, of size $q^{f_i} \le q^N$, and use the unique rank-gap test on each basis vector of the current parameter space. There are polynomially

many coordinate tests. The tests for (10)–(11) use all coefficient vectors and exact integer comparisons, giving the stated bound. This field-coordinate procedure is not an enumeration of all multiplicity matrices at once. □

3.3 Canonical successive value blocks

Corollary 3.5. *Suppose an actual group H normalizes E and acts compatibly on V. The value extraction in Theorem 3.4 can be made covariant under H as a succession of linear maps on their actual kernel domains. Its value spaces carry a computed H-action trivial on $\mathrm{U}(E)$. In the scalar-prefix convention, each successive value-map transporter is a genuine-bimodule instance, and its exact pure stabilizer is a nested star-subalgebra. The final kernel is the literal low-rank kernel; no later functional is extended arbitrarily to an earlier domain.*

Proof. Use the canonical radical Loewy flags of A and V/A, concatenated to a flag of V containing A. They are preserved by H, their successive quotients are semisimple E-modules, and $1 + J(E)$ acts identically on each quotient. Within a quotient take the sum of its large isotypic parts. That sum is also normalizer-invariant. For output layer a and input layer b, use the entire value space

$$\mathcal{C}_{ab} = \mathrm{Hom}_{E/J(E)}\big((\mathrm{gr}_b V)_{\mathrm{large}}, (\mathrm{gr}_a V)_{\mathrm{large}}\big).$$

Each nonzero element has rank greater than τ, since its image contains a large simple module. Replace each individual-copy extraction by the whole block in $\mathcal{C}_{ab}$. The single error in (14) still has rank less than $2r$ after restriction and projection. Thus the same uniqueness and linearity argument applies to the whole block.

Here is the normalizer covariance explicitly. An element $h \in H$ preserves the Loewy flag. Conjugate its image of L back to L by an element of $1 + J(E)$. The remaining normalizer of L is upper block triangular on the split flag and intertwines the same simple-factor action on all copies. Its unitriangular part preserves the previously zero output/input pattern; its diagonal blocks act naturally on $\mathcal{C}_{ab}$. Conjugating the entire decomposition (14), rather than separately bounding many block errors, preserves the one rank bound. Uniqueness now proves that the next extracted value transforms by those diagonal blocks. A change of complement or split coordinates is treated by the same argument. This proves covariance on the current kernel domain.

An element of $\mathrm{U}(E)$ acts on the two layers through the same simple algebra matrices. A map in $\mathcal{C}_{ab}$ intertwines them, so this action is trivial. The value action therefore factors through $H/\mathrm{U}(E)$. Coordinates and its action are computed on the associated graded modules by linear algebra; temporary individual-copy coordinates may be used to recover a whole block, but are not declared canonical.

Write the resulting sequence as

$$K_0 = Z, \qquad \phi_j : K_{j-1} \longrightarrow \mathcal{C}_j, \qquad K_j = \ker \phi_j.$$

After the preceding maps are matched, their kernels are common invariant domains for the current algebra. With coefficient action $\rho(g) = \chi(g)^{-1}\pi(g)$, the value-map action is

$$\phi \longmapsto \chi(g)\, \sigma_{\mathcal{C}}(g)\, \phi\, \pi(g)^{-1}.$$

On the pure group it is $\chi(u)\phi\pi(u)^{-1}$, the conjugation action of the genuine bimodule with left scalar character χ and right coefficient module. Its exact stabilizer algebra is given by $\phi\pi(a) = \chi(a)\phi$ and the star equations. This is a unital star-subalgebra and preserves the new kernel. The compatible transporter theorem matches this one map with the full stabilizer before the next map is considered. In particular no moving value or kernel domain is silently frozen. □

Remark 3.6. The successive flag only makes the normalized pure action on Z/K_{last} unitriangular. It need not be trivial. Before using that entire quotient as fixed multiplicity, pass to

$$E_0 = \{a \in E : \pi_{Z/K_{\mathrm{last}}}(a) = \chi(a)I\},$$

with the star equations. The current group normalizes E_0, and the kernel of the normalized coefficient action of $\mathrm{U}(E)$ is exactly $\mathrm{U}(E_0)$. Its image is a p-group. Hence this passage preserves a unitary-by-p state without a branch. The pure coefficient action is now scalar on the full quotient, so its character cancels the permanent tensor-value character in the subsequent one-arm restriction.

3.4 A canonical physical rank cover over every base field

Lemma 3.7. *Let $\mathcal{S} \subseteq \mathrm{Hom}_{\mathbb{F}_q}(X, Y)$ be a matrix space all of whose BASE-FIELD elements have rank at most R. There are subspaces $U \subseteq X$ and $B \subseteq Y$ such that*

$$\mathcal{S}U \subseteq B, \qquad \operatorname{codim}_X U + \dim B \le 4R.$$

No lower bound on q is required.

Proof. Choose a base-field $A \in \mathcal{S}$ of maximum rank a, and use coordinates in which A has an a-dimensional identity block and zeros elsewhere. For $B \in \mathcal{S}$, let b be the rank of its complementary lower-right block. Reduce that block to an identity of size b. Its row and column pivots can clear the adjoining blocks of B without changing A, since the corresponding rows and columns of A vanish. Thus the two matrices B and $A + B$ have a common separate identity block of size b, and their remaining parts differ by a rank-a matrix. The rank triangle inequality gives

$$\operatorname{rank} B + \operatorname{rank}(A + B) \ge 2b + a.$$

Both matrices are base-field elements of $\mathcal{S}$, so $b \le a/2$. In particular the complementary projected matrix space has maximum base-field rank at most $a/2$.

Delete the a pivot-domain directions and allow the a pivot-image directions in the cover, then repeat on that projected space. The successive maximum ranks are at most $R, R/2, R/4, \ldots$. At termination the remaining projected space is zero. The sum of deleted domain and allowed image dimensions is at most $2\sum_{j\ge 0} R/2^j = 4R$. This argument only uses A, B, and $A + B$ over the original field. □

Proposition 3.8. *There is a deterministic canonical choice of a pair $(U, \mathcal{S}U)$ with*

$$\operatorname{codim}_X U + \dim \mathcal{S}U \le 4R.$$

For the physical spaces and pencils in the present algorithm it is computable in $q^{O(N)}\,\mathrm{poly}(\log q)$ time. More generally the construction has only polynomial dependence on matrix dimensions, apart from a conservative $q^{O(1)}$ finite-field factor.

Proof. Write ρ for noncommutative rank. Every cover in Lemma 3.7 bounds the rank of every blow-up $\mathcal{S} \otimes \mathrm{Mat}_e(\mathbb{F}_q)$ by $4Re$: on $U \otimes \mathbb{F}_q^e$ its image lies in $B \otimes \mathbb{F}_q^e$, and the complementary domain has dimension $e \operatorname{codim} U$. Hence $\rho \le 4R$.

The deterministic noncommutative-rank algorithm [7, Theorem 1.5 and Corollary 1.7(2)] supplies an optimal-rank BASE-FIELD blow-up

$$B \in \mathcal{S} \otimes \mathrm{Mat}_e(\mathbb{F}_q), \qquad \operatorname{rank} B = e\rho.$$

Its small-field construction passes through an extension and then uses the regular representation to return a base-field blow-up; an arbitrary extension-field subspace is not used as the output here. The standard shrunk-space characterization gives minimum cover cost ρ.

For completeness, canonicity follows by the following Wong sequence in the BLOW-UP domain. After zero-padding rectangular spaces to a common size d, put $\mathcal{S}_e = \mathcal{S} \otimes \mathrm{Mat}_e(\mathbb{F}_q)$ and

$$U_0 = \ker B, \qquad U_{j+1} = B^{-1}(\mathcal{S}_e U_j).$$

If U has maximum deficiency $e(d-\rho)$, then

$$e(d-\rho) \le \dim(U \cap \ker B) \le \dim \ker B = e(d-\rho).$$

Thus $\ker B \subseteq U$ and $BU = \mathcal{S}_e U$, whence $B^{-1}(\mathcal{S}_e U) = U$. Every sequence term is consequently contained in every maximum-deficiency subspace. The increasing sequence stabilizes in polynomially many steps. At its limit its image lies in $\operatorname{im} B$, because it is contained in a maximum-deficiency subspace, and $BU_\infty = \mathcal{S}_e U_\infty$. Its deficiency is therefore $\dim \ker B$, so it is itself maximum and is the unique smallest such space.

Right changes of basis on the blow-up index preserve $\mathcal{S}_e$. Uniqueness makes U_∞ invariant under all $I_X \otimes \mathrm{GL}_e(\mathbb{F}_q)$. These matrices span $I_X \otimes \mathrm{Mat}_e(\mathbb{F}_q)$ in odd characteristic, giving $U_\infty = U_* \otimes \mathbb{F}_q^e$. Recover U_* by the linear equations $x \otimes e_1 \in U_\infty$. It is the unique smallest original maximum-deficiency subspace. If padding added a zero domain summand, that summand belongs to every maximum-deficiency space and is discarded after descent. The image $\mathcal{S}U_*$ is computed by linear algebra. The resulting original pair is therefore covariant under every left-right equivalence, independently of the blow-up witness chosen.

The finite-field algorithms in the cited theorem are used with their deterministic finite-field implementation. It suffices here to retain a possible polynomial factor in p or q, rather than assert an unjustified polynomial dependence on $\log q$ alone. A small field is extended only to size polynomial in the matrix dimension, and direct deterministic field construction has that same bound. These costs are absorbed in $q^{O(N)}$ on the physical modules. The rank R in the statement is the maximum rank over the ORIGINAL field; no large-field commutative-rank identity is assumed. □

In a bulk step one arm suffices. Apply Proposition 3.8 to the maps $A_j \to T_k^*$ given by the literal kernel pencil. If its canonical pair is (U, B), put $U_k = A_k \cap \operatorname{ann}(B)$. Then

$$F(K, U, U_k) = 0, \qquad \operatorname{codim}_{A_j} U + \operatorname{codim}_{A_k} U_k \le 4R.$$

This is a pair of actual physical subspaces, and all its constructions are covariant. If $R < r$ and $\tau = 10r$, its quotient degrees are less than τ. After its actual transporter is imposed, let I' be the new cutoff ideal. The identity $I'T_j = I'(I'T_j) \subseteq I'A_j \subseteq I'T_j$ shows $I'T_j = I'A_j$; it is a MODULE identity, not an assertion that the new ideal is an ideal of the older algebra. Since A_j/U is too small to support a new large simple module, $I'A_j \subseteq U$. The analogous statement holds in the other row. Trimming the two rows to $I'T_j, I'T_k$ therefore preserves the literal zero block and exposes only small natural-degree boundary types.

3.5 A finite total charge for outer labels and field actions

Lemma 3.9. *Let $E_0 \supseteq E_1 \supseteq \cdots$ be actual unital algebras on the same original faithful degree-N module. Suppose the successive actual allowed groups are nested, normalize every retained earlier algebra, and fix its already imposed factor and center labels. Fixing all new factor labels and center-field actions by their actual quotient images costs at most $N!$ in product of indices when the initial*

centers are the base field. For general initial centers an additional one-time factor $\prod_i f_i \le \exp(O(N))$ *suffices.*

Proof. Every new simple factor has a nonempty set of old neighbors. Assign it to the least labeled old neighbor. The actual group preserves this assignment; no natural-copy frame or invariant complement is chosen. If $A = \mathrm{Mat}_a(K)$ is an old parent and its assigned children are $B_j = \mathrm{Mat}_{b_j}(L_j)$, let t_j be the K-rank of the identity of B_j in the old natural module. The idempotents are orthogonal, so $\sum_j t_j \le a$ and $b_j \le t_j$. Fixing the h child labels costs at most $h!$.

Assume the old center $K = \mathbb{F}_{q^f}$ is fixed. For a fixed child $B = \mathrm{Mat}_b(L)$, $L = \mathbb{F}_{q^g}$, in a corner of K-degree t, put $d = \gcd(f, g)$. The occupied components of $K \otimes_{\mathbb{F}_q} L$ each have natural K-degree $b(g/d)$. A center-automorphism subgroup induced by K-linear normalizers has kernel of order at most g/d on these component labels. The orbit of any occupied component has size at most the number of occupied components. Thus its order is at most

$$(g/d)\,\#\{\text{occupied components}\} \le t/b.$$

The normalizer must preserve multiplicities, so unequal multiplicities cannot enlarge this bound. Equivalently, over an algebraic closure these automorphisms act freely on the occurring center embeddings, each using at least b dimensions.

The joint field-action image after fixing labels has order at most the product of the child bounds. Therefore its combined local index is at most

$$h! \prod_j \frac{t_j}{b_j} \le h! \prod_j \frac{t_j!}{b_j!} \le \frac{a!}{\prod_j b_j!}.$$

The last inequality is the multinomial bound with the unused $a - \sum_j t_j$ positions included. Each new factor has just one assigned parent, even if it is a diagonal merger of several old factors. These assignments form a rooted forest, and the factorial ratios telescope to $\prod_{\text{roots}} a_i! \le N!$. Other old parents of a merged child simply retain unused budget; their labels remain imposed on the actual group. Finally $\sum_i f_i \le N$ gives the stated initial-center bound.

The quotient actions are computed by conjugation on the semisimple algebra data, with actual matrix words and coset lifts. Deterministic permutation-group operations in degree $q^{O(N)}$ suffice [6]. No unrestricted group-intersection computation, or normalization of an arbitrary source subspace, is part of this charge. □

Remark 3.10. The nested-group condition in Lemma 3.9 is essential. A later enlargement which discards old factor labels would invalidate the forest. All refinements in the physical-pass algorithm retain the earlier normalizers. Simultaneous source conjugation merely conjugates the whole stored forest and leaves its degrees and numerical charge unchanged.

4 A terminal for the entire supported tensor

The terminal in this section treats the full tensor in one orbit problem. No leading, middle, or corner coordinate is discarded.

Theorem 4.1 (Supported-tensor terminal)**.** *Let H preserve physical interiors $T_i \le V_i$, and let $\mathcal{Z}$ be the space of tensors vanishing on $T_1 \times T_2 \times T_3$. Allow a supplied tensor-value character $\theta : H \to k^\times$ multiplying ordinary physical pullback. Suppose a full normal subgroup $\mathrm{U}(E_0)$ is supplied, with solvable quotient, and E_0 acts by a common scalar λ on the full primal boundary $\bigoplus_i V_i/T_i$ and by its*

star partner on the paired dual. All normalizer and representation compatibility maps are supplied in faithful degree $O(N)$.

Then full transport in $\mathcal{Z}$, including the full source stabilizer, costs $q^{O(N)}\,\mathrm{poly}(L,\log q)$. The theorem also holds for a single supported arm.

Proof. Put $A_i = \mathrm{ann}(T_i) \le V_i^*$ and form

$$W = (A_1 \otimes V_2^* \otimes V_3^*) \oplus (V_1^* \otimes A_2 \otimes V_3^*) \oplus (V_1^* \otimes V_2^* \otimes A_3).$$

The sum of the three inclusions is a surjection $\pi : W \to \mathcal{Z}$. It is H-equivariant because every T_i is preserved. Keep its complete kernel P. In particular, a monomial belonging to two or three arms is not assigned an arbitrary preferred representation.

Let $\epsilon(u) = \lambda(u)^{-1}$ be the common character on the A_i. Adjoin a temporary central scalar s multiplying tensor values, and embed

$$\kappa : \mathrm{U}(E_0) \longrightarrow H \times k^\times, \qquad u \longmapsto \big(u, [\theta(u)\epsilon(u)]^{-1}\big).$$

The subgroup is normal: conjugation preserves both scalar characters. On each arm, the total character $\theta\epsilon$ cancels. What remains is the tensor product of two genuine row modules and a fixed multiplicity space. For one-sided actions ρ_1, ρ_2, define the bimodule action by

$$a \cdot (v \otimes w) \cdot b = \rho_1(a)v \otimes \rho_2(b^*)w.$$

For unitary u, $(u^{-1})^* = u$, so conjugation gives exactly the two required row actions. Thus W is a genuine E_0-bimodule for $\kappa(\mathrm{U}(E_0))$. Its kernel P is invariant under the entire actual group; it is not claimed to be an algebra-bimodule subspace.

The quotient $(H \times k^\times)/\kappa(\mathrm{U}(E_0))$ is solvable. Apply the compatible invariant-quotient transporter theorem (Theorem E.3) to W/P. Intersect the complete returned coset with $s = 1$ using the explicit character image. This gives exactly the requested full transporter under H. Any other required character can be imposed on that same complete coset. The temporary scalar has bounded faithful degree and is then discarded.

In the recurrence, $\theta = \chi$. For a one-arm residual with $\lambda = \chi$, cancellation already occurs and no temporary scalar is needed. The permanent χ is retained until the final original-tensor transporter is intersected with $\chi = 1$. □

Corollary 4.2 (Bounded boundary matrix sizes)**.** *Suppose the current group G contains full normal $\mathrm{U}(E)$ with p-group quotient, preserves the physical interiors, and acts compatibly. If every simple factor of the boundary image algebra has matrix size at most an absolute constant a_0, the full supported-tensor problem is solvable in $q^{O(N)}\,\mathrm{poly}(L,\log q)$ time.*

Proof. Let B be the image algebra on the boundary and its paired dual. If $B/\operatorname{rad} B = \prod_i \mathrm{Mat}_{a_i}(\mathbb{F}_{q^{f_i}})$, then

$$\sum_i f_i a_i^2 \le a_0 \sum_i f_i a_i = O(N).$$

Enumerate the actual conjugation image of G on $B/\operatorname{rad} B$, retaining original words. Its inner part has order at most $q^{\sum_i f_i a_i^2}$ and its field-automorphism part has at most $\prod_i f_i \le q^{O(N)}$ choices. Inner algebra-unitary automorphisms fix each simple factor, so the permutation image is a p-group on at most $O(N)$ factors. Its order is at most the p-part of the corresponding factorial, hence at most $p^{O(N)}$. The entire image therefore has $q^{O(N)}$ elements.

For each image coset normalize the whole target. Let K be the full kernel of this conjugation action, with all actual lifts retained. Let E_0 be the star-subalgebra acting by a common scalar on the *full* boundary. Then $\mathrm{U}(E_0) \le K$ and is normalized by K. Put $L = K \cap \mathrm{U}(E)$. The semisimple boundary image of L consists of central units, and its remaining boundary image is unipotent. Thus its boundary image is solvable. The kernel of the full boundary representation inside $\mathrm{U}(E)$ lies in $\mathrm{U}(E_0)$, so $L/\mathrm{U}(E_0)$ is solvable. Also K/L is a subgroup of the supplied p-group $G/\mathrm{U}(E)$. Hence $K/\mathrm{U}(E_0)$ is solvable.

Apply Theorem 4.1 with $H = K$ and unite the actual cosets. This is a full-preimage construction inside the supplied algebra-unitary state. It makes no assertion about an arbitrary subgroup of that state. Large commutative extension fields are allowed: only a_i, not $f_i a_i$, is bounded in this argument. □

5 The controlled physical pass

This section gives the actual state and operations. The group is stored by original matrices and words, not by an independent product of its images.

Definition 5.1 (Pass state)**.** A state consists of the two full tensors, a current actual transporter coset, physical interiors satisfying (2), and a normal pure kernel $\mathrm{U}(E)$ whose quotient in the source group H is a p-group. The normalizer action on E and every module action is explicit. The scalar anchor and weighted actions are those of Section 1. Every earlier actual row or coefficient flag, partial-form equation, value graph, and source normalization is retained. In particular, a later enlargement of an interior does not release an equation previously matched on a smaller or different row space.

The initial state has $H = \mathrm{U}(E_{\mathrm{init}})$ and trivial quotient. The following facts prove that the p-only condition is inductive, rather than requiring a new solvable-image enumeration at every operation.

Lemma 5.2 (Pure kernels and normalized socle copies)**.** *The normalizer reductions in Lemma 3.9 allow the current group, with actual coset lifts, to be written*

$$H = \mathrm{U}(E)C_p = (\mathrm{U}(1 + \operatorname{rad} E)C_p) \rtimes \mathrm{U}(F),$$

where F is an actual star-Levi and C_p is a p-group centralizing F. For a nonzero genuine coefficient quotient module, an actual H-invariant E-simple copy of natural degree δ can then be chosen in its socle. A vector in that copy has at most q^δ possible target images.

Proof. Fix factor labels and center actions with the charged actual lifts. Normalize the Levi using elements of the contained unitary radical. A remaining star-automorphism of a simple factor is inner; its obstruction to being unitarily inner is a norm or square class of order prime to p. The quotient is a p-group, so this obstruction is trivial. Correct each lift by an actual unitary element. The resulting centralizer supplement has central prime-to-p kernel $\mathrm{U}(Z(F))$ and p-group quotient. It is that central group times a normal p-group. The central part is absorbed into $\mathrm{U}(F)$. The normalizer calculations and the total factor/field indices are exactly those in Lemma 3.9; no fresh factorial budget is granted here.

The unitary radical acts trivially on $\mathrm{Soc}_E(C)$. The group C_p preserves each fixed isotypic component and commutes with F. A p-group in characteristic p has a nonzero common fixed space on every nonzero module. This fixed space is F-stable, so it contains an F-simple summand. It is an E-simple submodule of the socle and is preserved by the actual H. The scalar normalization in

(3) preserves the same copy. This proves the target-image bound without claiming that all abstract irreducible H-modules are E-simple. □

Fix direction i. During this pass only T_i may grow; T_j, T_k only shrink. Maintain a controlled subspace $L \le C_i$ on which the entire current restriction $Q_i(c) = T(c, T_j, T_k)$ has already been matched. Its coefficient image is a p-group under (3). The pure algebra satisfies, for every $a \in E$ and $s \in L$,

$$[a, M_s] = 0, \qquad \pi_i(a)s = \chi(a)s,$$

together with the star-partner equations. In particular its unitary kernel commutes with all controlled contractions and acts physically on L by χ. The full group preserves the controlled graph

$$P_L = \{(M_s, s) : s \in L\}. \tag{15}$$

It is fixed pointwise by the pure unitary kernel. Initially $L = 0$.

Choose a simple copy $S \le \mathrm{Soc}_E(C_i/L)$ by Lemma 5.2, and let $L_0 \le C_i$ be its full preimage. Write $\delta = \dim S$.

Lemma 5.3 (A quotient mark with its actual lift)**.** *For a source vector $h + L \in S$ and one proposed target vector $h' + L$, the full transporter for their contraction and coefficient-lift equation is a single invariant-quotient bimodule transporter. The new pure zero-shift kernel is the unitary group of a star-subalgebra of the old E, and the quotient of the full new stabilizer by that kernel is a p-group.*

Proof. Transport (M_h, h) to $(M'_{h'}, h')$ modulo (15) under the action (4). The coefficient summand is a genuine bimodule with the character χ on one algebra side; the form summand is the genuine row bimodule. The lower graph is invariant under H and fixed pointwise by $\mathrm{U}(E)$. Thus Theorem E.3 applies. Identity (4) says that the same lower vector is used for both the coefficient lift and the form shift.

The exact pure kernel is obtained by the linear equations

$$[a, M_h] = 0, \qquad (\pi_i(a) - \chi(a))h = 0 \tag{16}$$

and their star partners, together with all older equations. These define a unital star-subalgebra $E_h \le E$. The pure modulo-graph stabilizer has an additive shift homomorphism to P_L, with zero-shift kernel $\mathrm{U}(E_h)$. The full new quotient is an extension of this additive p-group by a subgroup of the old p-quotient. It normalizes E_h: conjugation changes the pair by an element of a space fixed pointwise by the old pure kernel. The full coset, not one of its witnesses, is retained.

For $s \in L$, both equations (16) are unchanged when h is replaced by $h + s$. This lift-independence is also valid in the star equations. It will permit an actual linear kernel in the bulk step. □

5.1 The selected case

Fix a large absolute constant C. Bounded $\delta < 2C$ is handled by the whole-image case below. For an active type otherwise choose a positive dyadic r comparable to δ/C, and put $\tau = 10r$. Test each quotient vector using the exact selection predicate of Theorem 3.4. Entropy is measured on the original faithful module. The partial ideal-image dimensions are measured on the intrinsic row subquotient

$$W_{\mathrm{pass}} = T_j \oplus T_k \oplus T_j^* \oplus T_k^*.$$

No partial form is zero-extended through an arbitrary physical complement.

If a vector is selected, enumerate its at most q^{δ} target quotient images and use Lemma 5.3. A successful branch enlarges L by that one quotient vector. Its whole restriction is matched, because the graph equation enforces the actual lift. Thus $\dim(C_i/L)$ decreases by one. Each selected mark pays a potential loss at least r, and $\delta = O(r)$. Source scans cost at most q^N times the algebra routines and do not multiply the target branching count.

5.2 The no-progress bulk operation

Suppose no vector of S is selected. Lift-independence in Lemma 5.3 shows that the same no-progress conditions hold for the entire *actual* pencil on L_0, including the already controlled lower vectors. Apply Theorem 3.4 and Corollary 3.5. They produce a successive flag

$$L_0 = K_0 \geq K_1 \geq \cdots \geq K_t = K, \qquad \phi_\ell : K_{\ell-1} \longrightarrow W_\ell, \qquad K_\ell = \ker \phi_\ell.$$

The final K is a literal subspace of the physical boundary, and its contractions have rank less than r on the large-ideal row arms. No section of a quotient by the lower contraction span is chosen.

All target structural tests are checked as well. They are covariant under the current actual group, so a failed target no-progress test proves that this branch cannot contain a true isomorphism. For the remaining targets, align each actual value map on its actual domain. The value representation is fixed by the pure kernel and has the supplied external p-action. Its full action is

$$\phi \longmapsto \chi(g)\sigma_W(g)\phi\pi_i(g)^{-1}.$$

This is a genuine bimodule action. Its pure stabilizer satisfies $\phi\pi_i(a) = \chi(a)\phi$ and the star equations, so it is a nested star-subalgebra. Matching one map aligns its kernel, making the next domain a genuine module for the next algebra. No frame is added for each flag level, and no conditional map is extended arbitrarily to all of L_0.

After the flag has been matched, the normalized action on L_0/K is unitriangular. Transfer this action to the p-quotient by replacing the pure algebra with

$$E_0 = \{a \in E : \pi_i(a)|_{L_0/K} = \chi(a)I\}, \tag{17}$$

including the star equations. This is a normalizer-stable star-subalgebra. The homomorphism $u \mapsto \chi(u)^{-1}\pi_i(u)|_{L_0/K}$ has p-group image and kernel $\mathrm{U}(E_0)$, so $H/\mathrm{U}(E_0)$ remains a p-group without branching. The quotient may have large dimension; solvability comes from the matched values, not from a dimension estimate.

Let $A_j = I_\tau(E)T_j$ and $A_k = I_\tau(E)T_k$ for the algebra used in the relative test. Use ONE arm of the literal kernel pencil, from A_j to T_k^*. Its maximum base-field rank is less than $2r$ (the relative bound in fact gives the stronger r bound in the displayed representation). Proposition 3.8 gives a canonical physical zero block of total boundary dimension less than $8r < \tau$. Intersect the second row subspace with A_k. This does not assume a rank bound for a doubled-arm pencil. Align the resulting two physical subspaces under the *actual* current group by Corollary E.4, after the normalized decomposition in Lemma 5.2. This costs $q^{O(N)}$ and returns one full coset. Its pure kernel is the old algebra intersected with the subspace conductor; its quotient is a subgroup of the old p-quotient.

Write E_{new} for the resulting pure algebra and $B_j \subseteq A_j$, $B_k \subseteq A_k$ for the aligned cover spaces. Choose, exactly,

$$T_j' = I_\tau(E_{\mathrm{new}})T_j, \qquad T_k' = I_\tau(E_{\mathrm{new}})T_k.$$

These spaces lie in the cover. Indeed, ideal monotonicity gives $I_\tau(E_{\mathrm{new}})T_j \subseteq A_j$, while $\dim(A_j/B_j) < \tau$ implies that the same ideal annihilates A_j/B_j. The large ideal is idempotent, so

$$I_\tau(E_{\mathrm{new}})T_j = I_\tau(E_{\mathrm{new}})^2T_j \subseteq B_j.$$

The other row is identical. The conductor makes B_j, B_k genuine current modules, so this argument uses their actual quotient dimensions. These canonical new interiors are preserved by the current normalizer. Consequently

$$T(K, T'_j, T'_k) = 0. \tag{18}$$

Promote the full preimage of K into T_i, and replace the two other interiors by T'_j, T'_k. Equation (18) proves the full interior zero-block condition. Former middle or cubic terms are included in the current full contraction; no term is discarded during promotion.

Put $L_{\mathrm{new}} = L_0/K$ in the new boundary C_i/K. Its residual restriction is well-defined only on T'_j, T'_k, and this is where it is matched. The pure kernel in (17) acts on its physical coefficient space by χ. In (1) the tensor-value χ cancels the dual coefficient χ^{-1}, leaving a genuine row bimodule. Theorem E.3, or the one-arm case of Theorem 4.1, returns *one complete transporter coset* for this entire residual restriction.

Its exact ordered-contraction kernel is computed by adding all those form equations to E_0. The full quotient remains p: its coefficient image is unitriangular and the previous quotient was p. Retain all older equations on their original domains as well. The pass invariant is restored with $L = L_{\mathrm{new}}$.

Lemma 5.4 (Bulk progress)**.** *The bulk operation eliminates the entire chosen quotient copy S from the uncontrolled coefficient dimension, even when K does not contain L.*

Proof. The exact identity is

$$\dim(C_i/K) - \dim(L_0/K) = \dim(C_i) - \dim(L_0) = \dim(C_i/L) - \dim S.$$

The remaining coefficient action on L_0/K is controlled because both its value flag and its whole residual restriction were matched. Low rank alone would not establish the conclusion. □

5.3 Inactive copies and completion of a direction

For a bounded matrix-size type $\mathrm{Mat}_a(\mathbb{F}_{q^f})$, the entire pure coefficient image on S has order at most $q^{fa^2+O(1)} = q^{O(\delta)}$. The normalized p-supplement fixes the chosen copy. Enumerate the whole coefficient image with actual lifts and match its lifted frame and all contractions in one graph-quotient call. This enlarges L by δ and adds only an additive p shift quotient. Bounded $\delta < 2C$ is covered by the same argument, since $q^{\delta^2} = q^{O(\delta)}$. The cost is charged to the eliminated coefficient dimension, not to δ independent arbitrary vector images.

Every selected operation decreases $\dim(C_i/L)$ by one, and every bulk or whole-image operation decreases it by δ. Thus a pass has polynomial depth and ends with the whole boundary in the controlled space. Its actual boundary action is solvable. In particular no large nonabelian classical pure factor remains in that boundary, although a large commutative field degree need not be small.

6 Degree rounds and the complete running time

Lemma 6.1 (Monotone images in a physical pass)**.** *Fix a cutoff τ and a pass in direction i. Every refinement has $E' \le E$ and $T'_j \le T_j$, $T'_k \le T_k$. Then*

$$I_\tau(E')T'_j \subseteq I_\tau(E)T_j, \qquad I_\tau(E')T'_k \subseteq I_\tau(E)T_k.$$

The corresponding ideal-image dimensions on the paired dual row modules cannot increase either.

Proof. The natural-degree ideals are monotone under represented algebra inclusion, so the primal inclusions follow immediately. Shrinking T_j induces the equivariant quotient $T_j^* \to (T_j')^*$; an ideal image maps onto the ideal image in that quotient. This proves the dimension assertion for the duals. The same applies to k.

Source normalizations are actual elements of the preceding normalizer and preserve its current interiors. They conjugate all retained data together, so the numeric quantities are unchanged before the next inclusion. A coefficient promotion enlarges only T_i, which does not occur among the charged row modules in this pass. □

Lemma 6.2 (One-pass branch cost)**.** *A physical pass has polynomial depth. Apart from the global normalizer index forest, the sum of logarithms to base q of its branching factors on every path is $O(N \log^2(N+2))$.*

Proof. For each dyadic r maintain the entropy potential of Theorem 3.4, with $\tau = 10r$, together with the actual partial ideal-image dimensions of W_{pass}. Its entropy contribution on the original faithful module is at most $2N \log(N+2)$, its distinct-large-factor term is at most N, and its image terms are $O(N)$. It is monotone through all refinements at that fixed cutoff by Lemma 6.1.

A selected mark at that cutoff decreases this potential by at least r. Its target branching has at most q^δ possibilities, with $\delta = O(r)$. Summing over the $O(\log(N+2))$ available dyadic cutoffs gives the stated conservative $O(N \log^2(N+2))$ selected-mark cost. This does not assume that ideal images on a newly grown row can be charged to their old images: rows do not grow during the charged interval.

Whole-image branches for inactive or bounded-degree copies cost $q^{O(\delta)}$ and remove δ from the uncontrolled coefficient dimension. Their total cost is $q^{O(N)}$ in a pass. A selected operation removes one uncontrolled dimension, and a bulk step removes δ by Lemma 5.4. Thus there are polynomially many operations. All nonbranching full-coset calls cost $q^{O(N)}$ each and contribute only a polynomial multiplicity to that exponential cost. □

Lemma 6.3 (Three passes decrease active degree)**.** *Let D be the largest active boundary degree at the beginning of a round. For a sufficiently large absolute choice of C in the pass algorithm, three passes, in directions $1, 2, 3$, leave largest active boundary degree at most D/C_0, where $C_0 > 1$ is an absolute constant.*

Proof. Every active quotient chosen in the round has natural degree at most D. An algebra restriction cannot increase the natural degree of a simple type: a nonzero representation of a child factor on an old natural module has dimension at least its own natural degree. Nor can an inactive parent produce an active matrix size. Promotions in the processed direction take quotients of its boundary.

Consider a bulk step. Its new row boundary is an extension of T_j/A_j, annihilated by the large ideal, and a quotient of the physical cover of dimension less than $8r$. The former has simple degrees at most $\tau = 10r$, and the latter has total dimension below τ. Subsequent algebra restriction preserves this bound. Choosing C large enough gives $\tau \le D/C_0$. The same statement holds in the other row. Inactive whole-image steps create no row boundary.

When a direction is finished, its whole boundary action is solvable, by the controlled-L loop. A sufficiently large classical unitary factor would give a nonsolvable image there, so no active pure factor remains. Later subgroups preserve solvability on that old quotient. They can add to its boundary only the cutoff-small row pieces just described. Extensions do not create new simple

composition types. After all three directions have been processed, every active boundary type is therefore at most D/C_0.

This argument does not bound the total new boundary dimension. It may be $\Theta(N)$. It also does not assert that a large commutative extension field has small natural degree: such a factor is inactive, and restrictions cannot turn it into a noncommutative factor of larger matrix size. □

Proof of Theorem 1.1. Initialize the exact weighted-action state from Section 1. Run the three physical passes repeatedly while the active degree is larger than a fixed absolute constant. Lemma 6.3 bounds the number of rounds by $O(\log(N+2))$. At the end all boundary matrix sizes are bounded, so Corollary 4.2 completes the full supported tensor. All old equations and their actual source groups have been retained, so the terminal is applied to the exact remaining state.

Each operation preserves every true transporter in at least one of its actual target branches. Quotient marks enumerate all images in the actual invariant simple copy. The graph equation enforces the same lift in both coordinates. Bulk maps and covers are covariant, and their entire residual restriction is matched by one full coset. Conversely every returned terminal element satisfies the full tensor equation, so broad auxiliary covers or repeated use of source constraints introduce no false final solution.

Lemma 6.2 gives total target-mark and whole-image path cost $O(N\log^3(N+2))$. Lemma 3.9 bounds the factor and center normalizer refinements by $(N!)^{O(1)}$, which is smaller than this bound. The old normalizers and labels are retained throughout, so this is one global forest budget, not one new factorial at every step.

For completeness this bounds the whole tree, including failed branches. At a branching node let b be its actual number of children. Choose each child uniformly. The probability of a leaf is the inverse product of the branching numbers on its path. The proved path budget bounds every such product by $q^{O(N\log^3(N+2))}$. Since the leaf probabilities sum to one, the number of leaves is at most this quantity. Polynomial depth bounds the total number of nodes. Every nonbranching primitive costs $q^{O(N)}\,\mathrm{poly}(L,\log q)$, so the total has the required bound.

All vector-space certificates remain polynomial in the original input. The row modules have original physical degree; old restrictions are fixed labels or explicitly stored quotient equations. The only frame groups are the two current physical-cover groups. Coinduction repairs inside the backend are local, are discarded after recovering actual words, and are not iterated as new input modules. Group generating lists are reduced in the original faithful representation. Straight-line programs share prior words rather than expanding them.

Some additive images are only $\mathbb{F}_p$-linear. The backend then restricts scalars, multiplying its faithful degree by $\log_p q$; its cost is $p^{O(N\log_p q)} = q^{O(N)}$. The global algebra, natural-degree potentials, and factorial forest are still computed over the original k and the original degree N. The exact pure kernels returned to the outer algorithm are recomputed from their k-linear form, graph, value, and subspace equations. No prime-field degree is inserted into the global logarithmic budget.

Unite all successful full weighted transporter cosets using actual source generators and representative ratios. Finally impose $\chi = 1$ on this full coset. Its representatives and generators now satisfy the original untwisted three-tensor equation. Since $N = O(n)$, the displayed bound in Theorem 1.1 follows. □

7 A fixed threshold in each degree round

The adaptive choice of the selection threshold used above is sufficient for Theorem 1.1. We now keep the same exact group state and change that choice: a single threshold is used throughout all three physical passes of a degree round. A simultaneous entropy potential pays for selections at every threshold without restarting its budget.

Lemma 7.1 (Simultaneous cutoff potential)**.** *Fix a geometric set*

$$\mathcal{T} = \{10 \cdot 2^j : 0 \le j \le \lceil \log_2(N+2) \rceil\}.$$

On the original faithful module W of dimension N, put

$$K_\tau(E) = |\mathcal{L}_\tau(E)|, \qquad \Omega(E) = 2\Phi(E,W) + \sum_{\tau \in \mathcal{T}} \tau K_\tau(E).$$

For every unital subalgebra $E' \subseteq E$ and every $\tau \in \mathcal{T}$,

$$\Omega(E) - \Omega(E') \;\ge\; \Psi_\tau(E,W) - \Psi_\tau(E',W) \;\ge\; 0. \tag{19}$$

Moreover $0 \le \Omega(E) \le 4N \log_2(N+2) + 2N$.

Proof. Use the corners from (8). For an old factor $A = \mathrm{Mat}_a(\mathbb{F}_{q^f})$, denote by t_B the rank over $\mathbb{F}_{q^f}$ of the corner belonging to a new factor B, and put $m_B = f t_B$. Thus

$$\sum_B t_B = a, \qquad \delta(B) \le m_B, \qquad D(B) \le f t_B^2.$$

The entropy loss contributed by one old natural module is exactly

$$\Delta_A = \sum_B m_B \log_2 \frac{f a^2}{D(B)}.$$

Choose B_0 with largest corner rank. Every other t_B is at most $a/2$, so

$$\Delta_A \ge 2 \sum_{B \ne B_0} m_B. \tag{20}$$

For each cutoff let $F_\tau(A)$ be the excess over one of the number of large children of A, with negative values replaced by zero. Natural degrees cannot increase under restriction, so an old factor which is not large has no large children. Whether or not B_0 is large, we have

$$F_\tau(A) \le \sum_{B \ne B_0} \mathbf{1}_{\delta(B) > \tau}.$$

Since $\sum_{\tau \in \mathcal{T},\, \tau < d} \tau < 2d$, it follows that

$$\sum_{\tau \in \mathcal{T}} \tau F_\tau(A) \le 2 \sum_{B \ne B_0} \delta(B) \le 2 \sum_{B \ne B_0} m_B \le \Delta_A.$$

Every old factor occurs in the faithful module at least once. The additional multiplicity contributions are nonnegative. Summing gives

$$\sum_{\tau \in \mathcal{T}} \tau F_\tau \le \Delta\Phi, \qquad F_\tau = \sum_A F_\tau(A), \tag{21}$$

where $\Delta\Phi = \Phi(E, W) - \Phi(E', W)$.

In the large-factor incidence graph, let G_τ count the excess old neighbors of new vertices, and let R_τ count old vertices having no new neighbor. Every new vertex has a neighbor, and edge counting yields

$$K_\tau(E') - K_\tau(E) = F_\tau - G_\tau - R_\tau.$$

Consequently

$$\begin{aligned}\Delta\Omega &= 2\Delta\Phi - \sum_{\sigma\in\mathcal{T}} \sigma F_\sigma + \sum_{\sigma\in\mathcal{T}} \sigma(G_\sigma + R_\sigma) \\ &\geq \Delta\Phi + \sum_{\sigma\in\mathcal{T}} \sigma(G_\sigma + R_\sigma) \\ &\geq \Delta\Phi + \tau(G_\tau + R_\tau - F_\tau) = \Delta\Psi_\tau.\end{aligned}$$

The last quantity is nonnegative by Proposition 3.2. Finally

$$\sum_{\tau\in\mathcal{T}} \tau K_\tau(E) = \sum_A \sum_{\substack{\tau\in\mathcal{T} \\ \tau<\delta(A)}} \tau < 2\sum_A \delta(A) \leq 2N,$$

and $\Phi(E, W) \leq 2N \log_2(N + 2)$. □

Corollary 7.2. *There is a deterministic algorithm for arbitrary three-tensor isomorphism over every finite field $\mathbb{F}_q$ of odd characteristic with running time*

$$q^{O(n\log(n+2))}\,\mathrm{poly}(\log q).$$

It returns an isomorphism when one exists and generators of the full source stabilizer. This is obtained by changing only the threshold rule in the algorithm of Theorem 1.1.

Proof. At the beginning of a degree round let D be the largest active boundary simple degree. If D is below a sufficiently large absolute constant, all remaining boundary matrix sizes are bounded, so use Corollary 4.2. Otherwise choose a dyadic integer

$$\frac{D}{2C} \leq r \leq \frac{D}{C}, \qquad \tau = 10r, \tag{22}$$

where C is a sufficiently large absolute constant. Keep this same r and τ for all three physical passes of the round.

Every active coefficient copy chosen during that round has degree $\delta \leq D$: algebra restriction cannot increase natural degree, promotions take boundary quotients, and a newly created row boundary has simple degrees at most $\tau < D$. Thus a selected quotient-vector mark has at most $q^\delta \leq q^{2Cr}$ target choices.

Theorem 3.4 does not require $r \leq \delta$ or $\tau < \delta$ for the coefficient copy being processed. Its hypotheses are the two no-progress inequalities for every member of the actual linear pencil. Its rank gaps concern only the *row* factors of natural degree greater than τ. It therefore still supplies the successive value flag and its literal low-rank kernel when a processed coefficient copy has degree smaller than τ. The controlled-space, graph-lift, flag-matching, cover, and residual-restriction steps are otherwise unchanged. In particular, there is no replacement of a full residual restriction by its leading part.

The all-field cover bound of Lemma 3.7 gives physical cover codimension $O(r) < \tau$, with the constants used in the pass. This holds for the actual base-field matrix-space rank and has no large-field hypothesis. The proof of Lemma 6.3 now applies with this single threshold: all boundary types introduced during the round have degree at most $\tau \le D/C_0$ for an absolute $C_0 > 1$. The largest active degree thus falls by a constant factor after the three passes. There are $O(\log(N+2))$ rounds. The separate whole-image branch for inactive copies of possibly large field degree is unchanged; these copies need not satisfy $\delta \le D$.

Charge a selected mark first to its entropy decrease if $\Delta\Psi_\tau \ge r$. Its target-choice exponent is $O(r)$. Lemma 7.1 pays for all such marks over the *entire* algorithm with $O(N \log(N+2))$. Every other algebra restriction also leaves Ω nonincreasing. Source conjugations preserve its value, and target normalizations do not change the source algebra or source potential. No entropy budget is restarted at a pass or a round.

A selected mark not paid in this way has partial ideal-image loss at least r. At the single fixed cutoff of one pass, Lemma 6.1 bounds the total such loss by $O(N)$. These marks therefore cost $q^{O(N)}$ per pass and $q^{O(N \log(N+2))}$ over all rounds. Whole-image branches remove controlled coefficient dimensions and likewise cost $q^{O(N)}$ per pass. The nested factor and center refinement forest costs at most $(N!)^{O(1)}$ globally by Lemma 3.9, also within the stated bound.

The exact state, polynomial depth, and full-coset primitives have not changed. The path budget is now $O(N \log(N+2))$. Applying the complete tree argument from the proof of Theorem 1.1 bounds all nodes, including failed target branches, by $q^{O(N \log(N+2))}$ times a polynomial factor. The $q^{O(N)} \operatorname{poly}(L, \log q)$ cost of each nonbranching primitive is absorbed. The same final union of full transporter cosets and the same single $\chi = 1$ restriction give the full original transporter and stabilizer. Since $N = O(n)$, the asserted bound follows. □

Remark 7.3. The threshold in (22) is determined by the degree at the *beginning of the round*. It is not reset to the degree of each successive coefficient copy. This distinction is what gives one partial-image budget per pass. The present corollary retains the logarithm arising from global algebra entropy and from the number of physical degree rounds; it does not establish a linear exponent.

A The semisimple central shift problem

We supply the semisimple result used by the main algorithm, including its treatment of large auxiliary bimodules.

Proposition A.1. *Let A be a semisimple $*$-algebra of faithful degree N over a finite field k of odd order. On a genuine A-bimodule B, conjugacy modulo $P \le B^A$ can be solved deterministically in $|k|^{O(N)}$ times polynomial auxiliary work. The algorithm lists every nonempty fixed-shift coset $u_p \operatorname{U}(A_v)$, with one representative per actual shift, where*

$$A_v = \{a \in A : av = va,\ a^*v = va^*\}.$$

It also returns generators of $\operatorname{U}(A_v)$ and the full modulo-P stabilizer.

A.1 Embedding a genuine bimodule

The enveloping algebra $A^e = A \otimes_k A^{\mathrm{op}}$ is semisimple. A basis of B gives an explicit surjection from a finite direct sum of copies of A^e onto B. Solve linear equations for an A^e-linear section, giving a

polynomial-size embedding into $(A^e)^d$. Choose the nondegenerate symmetric Frobenius functional

$$\tau(a) = \sum_i \mathrm{Tr}_{K_i/k}(\mathrm{tr}(a_i)), \qquad A = \prod_i \mathrm{Mat}_{a_i}(K_i).$$

The map

$$x \otimes y \longmapsto \big[z \longmapsto x\tau(yz)\big]$$

identifies A^e, as an A-bimodule, with $\mathrm{End}_k(A)$, with left and right actions given by composition with left multiplication. Place the d resulting operators in distinct $(j, 0)$ blocks on A^{d+1}. This embeds B into an endomorphism space, equivariantly for conjugation by $\mathrm{U}(A)$, and sends B^A into the commutant of A. All dimensions are polynomial. The controlling exponential degree remains the original N.

It suffices, therefore, to solve the problem for $M, M' \in \mathrm{End}_k(W)$ and $P \leq \mathrm{Cent}(\rho(A))$, with an explicit A-module W.

A.2 Enumerating all shifts

Write $A = \prod_i \mathrm{Mat}_{a_i}(K_i)$, with $K_i = \mathbb{F}_{q^{f_i}}$. On an active isotypic component use coordinates

$$W_i = K_i^{a_i} \otimes_{K_i} D_i,$$

where A_i acts on the first tensor factor. A commutant shift is zero between distinct isotypic components and is K_i-linear on W_i. Decompose an arbitrary k-linear diagonal block into Frobenius components

$$M_{ii} = \sum_{\sigma \in \mathrm{Gal}(K_i/k)} M_{ii}^{(\sigma)} \sigma.$$

This is a direct-sum decomposition computable by linear algebra. Conjugation preserves its summands. Expand the identity components as

$$M_{ii}^{(1)} = \sum_t A_{it} \otimes Q_{it}, \qquad M'^{(1)}_{ii} = \sum_t B_{it} \otimes Q_{it},$$

where the Q_{it} form a K_i-basis of $\mathrm{End}_{K_i}(D_i)$. A shift has the form $I \otimes \sum_t \nu_{it} Q_{it}$. The conjugacy condition forces

$$XA_{it} - B_{it}X = \nu_{it}X \quad \text{for all } t, \qquad X = u_i. \tag{23}$$

For operators $L_1, \ldots, L_s$ on a d-dimensional K-space, list all joint scalar weights by successive intersections

$$W_\lambda = W \cap \ker(L_j - \lambda), \qquad \lambda \in K.$$

No invariance of W under L_j is needed. At each stage the sum of the surviving dimensions is at most d, since distinct eigenvalues give independent spaces. Thus at most d joint weights survive. Applied to (23), there are at most a_i^2 weights on factor i. Enumerating field elements costs q^{f_i}, and

$$\sum_i f_i a_i \leq N, \qquad \prod_i a_i^2 \leq 4^N.$$

Hence every complete candidate shift can be enumerated within $q^{O(N)}$ and filtered by membership in P. Inactive factors impose no shift coordinates and remain in the subsequent algebra computation. All off-diagonal blocks and nonidentity Frobenius components are imposed together in the next step; their witnesses are not chosen independently.

A.3 Completing one fixed shift

For a candidate shift p put $y = w + p$ and compute

$$A_v = \{a : av = va,\ a^*v = va^*\}, \qquad X_y = \{c : cv = yc,\ c^*y = vc^*\}.$$

These equations may be imposed in the original bimodule or its faithful embedding. The space X_y is a right A_v-module. If it contains an A-unit c_0, then $X_y = c_0 A_v$: writing $c = c_0 a$ turns the two intertwining equations into the two centralizer equations. Deterministic module isomorphism tests $(A_v)_{A_v} \cong (X_y)_{A_v}$. If an isomorphism ϕ exists, take $c = \phi(1)$ and test whether c is a unit of A. This is conclusive: if any unit c_0 existed, every module generator would be $c_0 a$ for $a \in A_v^\times$ and would be a unit.

For an invertible c in X_y, the element $h = c^*c$ is selfadjoint and lies in A_v. The standard $*$-norm algorithm solves

$$d^*hd = 1, \qquad d \in A_v^\times,$$

or rejects. A successful $u_p = cd$ is unitary, and the complete fixed-shift coset is $u_p \,\mathrm{U}(A_v)$. The algebra A_v may have radical; the deterministic norm and unitary-generator routines apply to it. The module-isomorphism spaces have dimension at most $\dim A \le N^2$; large multiplicity spaces enter only the linear equations.

For self transport the shift map

$$u \longmapsto uvu^{-1} - v$$

is additive, because every unitary fixes P pointwise. Its kernel is $\mathrm{U}(A_v)$. Representatives for the distinct successful shifts together with this kernel generate the full stabilizer. The same representatives, or their ratios to one successful transporter, give the corresponding full transporter and source stabilizer for a distinct target. This proves Proposition A.1.

B Amortized semisimple branching

Let $A = \prod_i A_i$ be semisimple, with

$$A_i = \mathrm{Mat}_{a_i}(\mathbb{F}_{q^{f_i}}), \qquad D_i = f_i a_i^2, \quad \delta_i = f_i a_i.$$

Let H be the modulo-central-space stabilizer in the preceding leaf, let $\eta : H \to P$ be its additive shift map, and let $B = A_v$ be its exact source $*$-centralizer. For an active factor i, project the shift to its complete diagonal coefficient tuple, denoting the image by $\eta_i(H)$. Each weight space contains $u_i \pi_i(B)$ for a suitable unitary u, where $\pi_i : A \to A_i$ is projection. Distinct weight spaces are independent, so

$$|\eta_i(H)| \dim_k \pi_i(B) \le D_i. \tag{24}$$

For an inactive factor take a singleton shift image. The full shift is determined by its active tuples, and hence $|\eta(H)| \le \prod_i |\eta_i(H)|$.

Write

$$B/\operatorname{rad}(B) = \prod_j B_j, \qquad B_j = \mathrm{Mat}_{b_j}(\mathbb{F}_{q^{g_j}}), \qquad D'_j = g_j b_j^2, \quad \delta'_j = g_j b_j.$$

Choose a Levi complement of B and join i to j when B_j has nonzero image in A_i. Every vertex has a neighbor. For the neighbor set J_i, mutually annihilating simple images and restriction of the natural A_i-module give

$$\dim_k \pi_i(B) \ge \sum_{j \in J_i} D'_j, \qquad \sum_{j \in J_i} \delta'_j \le \delta_i, \qquad \sum_{j \in J_i} b_j \le a_i. \tag{25}$$

For the last inequality, the projected central idempotents have K_i-ranks adding to a_i. Every nonzero constituent of $\mathrm{Mat}_{b_j}(\mathbb{F}_{q^{g_j}})$ after scalar extension to K_i has K_i-dimension at least b_j. The radical of B can only increase $\dim \pi_i(B)$, so these inequalities do not discard it.

Define, for a simple algebra,

$$\mu\big(\mathrm{Mat}_a(\mathbb{F}_{q^f})\big) = \frac{8^{fa+a-2}}{fa^2}, \tag{26}$$

and define μ multiplicatively on semisimple products. A faithful degree-N representation gives $1 \le \mu(A) \le 8^{2N}$.

Lemma B.1. *Suppose positive integer parameters satisfy $\delta = fa$, $D = \delta a$, $\delta_j = g_j b_j$, $D_j = \delta_j b_j$, $\sum_j \delta_j \le \delta$, and $\sum_j b_j \le a$. Then*

$$\mu(\mathit{old}) \ge \frac{D}{\sum_j D_j} \prod_j \mu(\mathit{new}_j).$$

Proof. Put $d = \sum_j \delta_j$, $b = \sum_j b_j$, $t = \#\{j\}$, and $h = (\delta - d) + (a - b)$. The inequalities $\delta \le 2^{\delta - d} d$ and $a \le 2^{a-b} b$ give $D^2 \le 4^h (db)^2$. Weighted Cauchy–Schwarz gives

$$d^2 = \Big(\sum_j g_j b_j\Big)^2 \le \Big(\sum_j D_j\Big)\Big(\sum_j g_j\Big).$$

For positive integers $x_1, \dots, x_t$, $\sum_j x_j \le 2^{t-1} \prod_j x_j$. Apply this to the g_j and twice to the b_j to obtain

$$(db)^2 \le 8^{t-1}\Big(\sum_j D_j\Big) \prod_j D_j.$$

Therefore

$$D^2 \le 8^{h+2t-2}\Big(\sum_j D_j\Big) \prod_j D_j.$$

On the other hand, $\mu(\mathrm{old}) / \prod_j \mu(\mathrm{new}_j) = 8^{h+2t-2} \prod_j D_j / D$, proving the claim. □

Apply Lemma B.1 to every old factor using (25), and then use (24). Since every new vertex has positive degree and every simple potential is at least one, we obtain

$$\mu(A) \ge |\eta(H)| \mu(B/\operatorname{rad}(B)). \tag{27}$$

For a nonempty transporter, its distinct fixed shifts form a coset of $\eta(H)$. Thus their number is exactly $|\eta(H)|$, and all their source kernel algebras are B. Equation (27) is therefore a bound on the sum of child potentials in a branching tree, not just on a product along one path. This proves the fixed-shift branch bound used in the semisimple affine leaf and the augmentation algorithms.

C The semisimple affine subquotient leaf

This appendix repairs a group-invariant subquotient before using the central shift leaf. It applies to arbitrary cocycles, including those obtained from nonsplit prefix stabilizers.

Theorem C.1. *Let F be a semisimple star-algebra faithfully represented in degree N, and put $G = \mathrm{U}(F)$. Let Q be an explicit G-invariant subquotient of a genuine F-bimodule. Given a cocycle $z : G \to Q$ and $c \in Q$, the affine transporter from 0 to c can be computed deterministically in $q^{O(N)} \operatorname{poly}(L, \log q)$ time. It is returned as fixed-shift cosets $g_j \mathrm{U}(F')$, with a common source star-subalgebra $F' \le F$ and actual generators. For a potential ν satisfying $1 \le \nu(F) \le q^{O(N)}$, the sum of the child potentials is at most $\nu(F)$. If $F' = F$, there is at most one coset and the affine source is fixed by all of G.*

C.1 The universal degree-two image

Let W be the supplied faithful F-module. Every simple F-module occurs in W, and every simple genuine F-bimodule occurs in $V_0 = \mathrm{End}_k(W)$. Since $F \otimes F^{\mathrm{op}}$ is semisimple, every genuine bimodule embeds into a direct sum of copies of V_0. Let

$$A = \mathrm{span}_k\{\mathrm{Ad}(g)|_{V_0} : g \in G\}.$$

Then $\dim A \leq N^4$, and every invariant subquotient under consideration is an A-module. The action is induced by actual group words and their linear relations. We do not assert an action of $F \otimes F^{\mathrm{op}}$ on an arbitrary invariant quotient.

Here is the degree-two structure of A used below. After the center fields are decomposed, its modules are natural–dual tensors, including their relative Frobenius twists. Different classical factors give external irreducible tensors. In an exchanged general-linear factor, an equal-twist term is either an adjoint module or a natural square; the latter splits as symmetric and exterior squares in odd characteristic. Distinct twists are the corresponding two-digit irreducible tensors. Orthogonal and symplectic diagonal terms split by the form adjoint. A second-kind unitary factor has the adjoint term with its fixed-field form, a natural-square term at the involution twist, and the other two-digit terms. The finite-group irreducibility of the degree-two hearts and their distinctness are the degree-two statements in [8, Section 3.2 and the defining-characteristic proofs of Theorems 2.9 and 2.11]; the twisted terms use the Steinberg tensor-product description. Appendix D supplies the needed root calculations and cohomology without assuming that a heart has zero first cohomology.

After bounded-rank factors have been removed as described below, the only nonsemisimple terms are scalar–heart–scalar adjoint chains. Their extensions are actual nonsplit extensions for the finite group. One direct verification is useful. In the relevant symmetric, alternating, or Hermitian matrix component, the invariant trace pairing is nondegenerate. If its large simple heart split off as a submodule, its dimension, more than half that of the component, would force a nondegenerate restriction of the pairing. Its two-dimensional orthogonal complement over the appropriate center field would have only trivial composition factors. In odd characteristic a two-dimensional orthogonal group has prime-to-p order, so that complement would be trivial. This contradicts the single invariant scalar line. Duality gives the corresponding head. This argument is made over the appropriate center field before restriction of scalars.

The coefficient field of each heart is the full defining field of its chain, not an unspecified subfield. Here is a direct verification. For a general linear factor use $\mathrm{diag}(s, r, 1, \ldots, 1)$, where $s \in K^\times$ and $r = \pm 1$. The difference between its two traces on the adjoint heart is

$$2(s + s^{-1}) + 4(a - 2).$$

The same formula holds on a unitary adjoint heart over the fixed field K_0, now taking s in the norm-one subgroup of $K^\times$. For orthogonal or symplectic factors use two hyperbolic pairs with natural eigenvalues s, s^{-1}, r, r^{-1} and all remaining eigenvalues one. The difference on either degree-two heart is

$$4(s + s^{-1} + a - 4).$$

Subtracting the trivial composition factors does not change these trace differences. The set of $s + s^{-1}$ has at least $(|K| - 1)/2$ values in the first-kind cases, and at least $(|K_0| + 1)/2$ in the unitary case. If the field is a proper extension of k, these cardinalities exceed the order of every proper subfield. The ordinary defining-field traces therefore generate the full chain field over k.

To conclude, the k-image algebra on the absolutely irreducible heart has full matrix algebra as its defining-field span. Its radical must vanish, and it has one simple factor: a radical or additional

central idempotent would persist in that full matrix span. It is a matrix algebra over a subfield of the chain field. Its ordinary matrix traces belong to that subfield, so the preceding calculation makes the subfield equal to the whole chain field. This also explains why field traces down to k, which could erase information, are not used in this step.

For one such chain, write this coefficient field as K_i/k and its heart dimension as h_i. In scalar–heart–scalar coordinates the actual image is

$$\begin{pmatrix} \alpha & b_i & \gamma_i \\ 0 & D_i & e_i \\ 0 & 0 & \alpha \end{pmatrix}, \qquad \alpha \in k, \quad b_i, D_i, e_i, \gamma_i \text{ over } K_i. \tag{28}$$

Indeed the semisimple quotient is $k \times \mathrm{Mat}_{h_i}(K_i)$. Each nonsplit adjacent extension forces a nonzero radical row or column bimodule; these are simple, hence full. Their products give the whole K_i top-to-bottom coordinate. There can be no other entries because the invariant flag and the two identical trivial diagonal characters are preserved. This also explains why replacing the chain by its heart would be invalid.

All repeated, dual, and Frobenius-conjugate occurrences of the same heart type are grouped first. In particular the two adjoint occurrences of an exchange pair, and the two fixed-field forms in a unitary matrix algebra, are not treated as independent group factors. The remaining distinct chain types have independent supporting classical factors. They share one trivial simple character. On their sum the image is therefore

$$A_0 = k1 + \bigoplus_i R_i, \tag{29}$$

where R_i is the zero-scalar part of (28). The differences $g_i - 1$ supported on a single classical factor generate its full augmentation ideal; the row and column products just described give all its radical entries. Products of ideals for different chain types vanish. All other simple degree-two types are separate semisimple blocks. These descriptions are effective in the original center coordinates; no generic algebra search in faithful degree N^2 is used.

C.2 An explicit dual regular embedding

Let V_i denote the whole chain in its genuine matrix-bimodule component. Let e_0 select the common trivial vertex in (29), and let e_{ij} select the jth heart coordinate. The right ideal e_0A_0 has coordinates $\alpha \in k$, $b_i \in K_i^{h_i}$, and $\gamma_i \in K_i$. For $v_i = (u_i, y_i, z_i) \in V_i$, define

$$P_* = \{(u_i, 0, 0)_i : \sum_i \mathrm{Tr}_{K_i/k}(u_i) = 0\}.$$

It is a central subspace of the sum of the genuine matrix bimodules. The pairing

$$\langle a, v \rangle = \sum_i \mathrm{Tr}_{K_i/k}(\text{first scalar coordinate of } av_i)$$

identifies

$$(e_0A_0)^* \cong (\bigoplus_i V_i)/P_*, \qquad (e_{ij}A_0)^* \cong V_i/(K_iI). \tag{30}$$

The free row and top-to-bottom entries detect the heart and top coordinates by the nondegenerate field trace pairing. The common scalar detects exactly the remaining scalar quotient. Associativity of the pairing proves equivariance. This argument does not use $\mathrm{Tr}(1) \neq 0$; trace remains nonzero and

nondegenerate when the extension degree is divisible by p. Both sides of the resulting decomposition of A_0^* have dimension

$$1 + \sum_i [K_i : k](h_i + 1)^2.$$

The other semisimple blocks of A^* embed in copies of their already present degree-two modules. A remaining trivial block embeds in a central matrix line. We obtain an explicit polynomial-size embedding

$$A^* \hookrightarrow B_*/P_*, \qquad B_* \text{ a genuine } F\text{-bimodule}, \quad P_* \le B_*^F. \tag{31}$$

For an arbitrary A-module Q, use coinduction:

$$Q \hookrightarrow \mathrm{Hom}_k(A, Q), \qquad v \longmapsto [x \mapsto xv], \qquad (af)(x) = f(xa).$$

Evaluation at 1 proves injectivity. Tensoring (31) with the underlying vector space of Q yields

$$j : Q \hookrightarrow B/P_0, \qquad P_0 \le B^F, \tag{32}$$

with polynomial auxiliary dimensions. A supplied action commuting with G extends by postcomposition on $\mathrm{Hom}_k(A, Q)$, hence acts only on the original Q multiplicity coordinate in (32). It preserves P_0 and retains every actual group correlation.

All matrix and field coordinates here are computed by polynomial-dimensional linear algebra in the explicitly described components. Finite-field factorization can use deterministic enumeration over an original center field of degree at most N, with polynomial extension degree, within $q^{O(N)}$ work. No enumeration of $q^{\dim A}$ or $q^{\dim Q}$ occurs.

C.3 Small ranks and the tree potential

For a bounded natural matrix size $2 \le a \le 12$, enumerate the unitary factor modulo its algebra-central unitary group and replace that algebra factor by its center. This is an actual coset decomposition. Its center has prime-to-p unit group. The construction above and the large-classical cohomology lemma apply to the reduced algebra, not to the unreduced small group. In particular, no affine coboundary assertion is made for $\mathrm{SL}_2(3)$.

Let $\delta(F)$ be the sum of the minimal faithful degrees of its simple algebra factors. Define

$$\nu(F) = \mu(F) q^{40\delta(F)},$$

where μ is the multiplicative potential in Appendix B. Replacing $\mathrm{Mat}_a(\mathbb{F}_{q^f})$ by its center decreases δ by $f(a-1)$ and has unitary index at most q^{fa^2}. The fixed constant 40 pays every bounded-rank index, including exchange pairs. The semisimple faithful degree cannot increase under restriction. The fixed-shift μ inequality therefore also contracts ν. Hence the sum of child potentials is at most the parent potential and $1 \le \nu(F) \le q^{O(N)}$.

For every bounded-rank center coset, normalize only the target by the full affine action. Keep source 0 and the same restricted cocycle. Thus the source embedding below and its exact source algebra are shared by all target-normalized cosets. If a noncommutative factor was replaced, that source algebra is proper in the old algebra; it is not a no-loss step.

C.4 Linearizing the affine cocycle

By Lemma D.1, the whole genuine-bimodule ambient space B satisfies $H^1(G, B) = 0$, and the central space P_0 satisfies $H^2(G, P_0) = 0$, after the bounded-rank reduction. The exact sequence gives

$$H^1(G, B/P_0) = 0.$$

Given the original cocycle $z : G \to Q$, solve on generators the linear equations

$$j(z(g)) = gb - b \quad \text{in } B/P_0. \tag{33}$$

They are consistent by the vanishing just proved. The map $x \mapsto j(x) + b$ converts the original affine action into ordinary unitary conjugation on the ambient central quotient. Apply Proposition A.1. It supplies every nonempty fixed-shift coset, one common exact source star-algebra F', and the fixed-shift branch bound. Together with the bounded-rank charge, this proves Theorem C.1. The equation is imposed in the original actual group through its generators, so the result includes witness words and full stabilizers.

For clarity, this applies to an arbitrary affine cocycle, not merely a translated fiber already embedded in the original module. If a commuting solvable group also acts affinely, with linear part c and shift $z_C(c)$, extend its linear action functorially as above and set

$$d_C(c) = j(z_C(c)) + b - cb.$$

Affine commutation implies that $d_C(c)$ is G-fixed on the whole ambient quotient, and its cocycle identity is inherited from that of z_C. Homogenize this external affine action by one fixed coordinate and apply the linear repair once more. Two local repairs still have polynomial size.

In every augmentation recursion the grade is represented as a subquotient of the *original* ambient bimodule. The local coinduction repair is discarded after recovering original group words. Repeatedly feeding an expanded repair into the next repair is not part of the algorithm.

D Cohomology of whole degree-two classical modules

This appendix proves the cohomological input for the affine semisimple transporter. It concerns whole Hom modules and specified quotients, rather than arbitrary simple constituents of their tensor squares. We use the convention

$$z(gh) = z(g) + gz(h), \qquad z(g) = (g - 1)b$$

for cocycles and coboundaries. Scalar extension commutes with these finite linear equations. We may therefore work over an algebraic closure and subsequently descend every required solution by linear algebra.

Lemma D.1. *Let F be a semisimple algebra with involution over a finite field of odd characteristic. Suppose each noncommutative matrix factor has matrix size greater than 12 over its defining field; commutative factors are allowed. Put $G = U(F)$. If B is a genuine F-bimodule with action $b \mapsto ubu^{-1}$, then*

$$H^1(G, B) = 0.$$

For every trivial characteristic-p coefficient space P, $H^1(G, P) = H^2(G, P) = 0$. In particular, for a G-invariant central subspace $P \subseteq B$,

$$H^1(G, B/P) = 0.$$

All statements permit restriction of scalars to $\mathbb{F}_p$. A supplied cocycle in the last module is made a coboundary by solving linear equations on the supplied group generators.

D.1 The special linear calculation, including the small field

Let $K = \mathbb{F}_{p^e}$, $G = \mathrm{SL}_r(K)$ with $r \geq 5$, and let A_i denote the natural representation with entries twisted by p^i, $0 \leq i < e$. We first prove

$$H^1(G, \mathrm{Hom}(U, V)) = 0, \quad U, V \in \{1, A_i, A_i^* : 0 \leq i < e\}. \tag{34}$$

The diagonal determinant-one torus T has p-prime order. Subtract a coboundary so that $z|_T = 0$. Write $x_{ab}(s) = I + sE_{ab}$. Then

$$z(tx_{ab}(s)t^{-1}) = tz(x_{ab}(s)).$$

Consequently a root value lies in the fixed space of $\ker(t \mapsto t_a/t_b)$.

Characters of T are exponent vectors modulo $|K| - 1$ and the all-ones vector. A weight supported on at most two coordinates is trivial on that root kernel only if it is a multiple of $\varepsilon_a - \varepsilon_b$ modulo $|K| - 1$. Indeed the root character is onto, and $r \geq 5$ leaves a coordinate outside both supports, forcing the all-ones adjustment to vanish. The natural and dual weights admit no such value. The weights $p^i\varepsilon_u - p^j\varepsilon_v$ with $i \neq j$ admit none either. For $A_i \otimes A_j$ or its dual, a surviving value would require $p^i + p^j = 0$ modulo $|K| - 1$, which occurs in odd characteristic only when $K = \mathbb{F}_3$ and $i = j = 0$. Thus only the equal-twist endomorphism module and this tensor square require further work. The trivial module has zero first cohomology because the root commutator relation makes $\mathrm{SL}_r(K)$ perfect.

For the endomorphism module, a Frobenius automorphism reduces to untwisted conjugation. Torus normalization gives

$$z(x_{ab}(s)) = D(s) + B(s)E_{ab} + C(s)E_{ba},$$

with D diagonal. The root cocycle identity makes C additive, and comparison in the two orders of parameters gives $sC(t) = tC(s)$. Hence $C(s) = cs$. Torus conjugation gives $C(\lambda s) = \lambda^{-1}C(s)$. For $|K| > 3$ a value $\lambda^2 \neq 1$ forces $c = 0$. For $K = \mathbb{F}_3$, use the order-three relation instead:

$$(1 + \mathrm{Ad}\, x_{ab}(1) + \mathrm{Ad}\, x_{ab}(1)^2)E_{ba} = E_{ab},$$

whereas this norm operator kills the diagonal space and E_{ab}. It again forces $c = 0$.

Now D is additive. Conjugating by a torus element with root value -1 fixes D, whereas additivity negates it, so $D = 0$. The remaining identities are

$$B(s + t) = B(s) + B(t), \qquad B(\lambda s) = \lambda B(s),$$

and therefore $B(s) = b_{ab}s$. Equivalently the cocycle defines the dual-number representation

$$\rho_\epsilon(g) = (I + \epsilon z(g))g, \qquad \epsilon^2 = 0,$$

which replaces the root parameter s by $s(1 + \epsilon b_{ab})$. Applying the relation $[x_{ab}(s), x_{bc}(t)] = x_{ac}(st)$ gives $b_{ac} = b_{ab} + b_{bc}$ for distinct indices. Thus $b_{ab} = d_b - d_a$ for scalars d_a, and the cocycle is the coboundary of $\mathrm{diag}(d_1, \ldots, d_r)$. This includes $p \mid r$ and $K = \mathbb{F}_3$.

It remains to examine $A \otimes A$ over $\mathbb{F}_3$. Odd characteristic splits it into symmetric and exterior squares. In the symmetric-matrix model $X \mapsto gXg^T$, write

$$z(x_{ab}(1)) = D + bS_{ab}, \qquad S_{ab} = E_{ab} + E_{ba}.$$

These are all root-kernel fixed vectors; all diagonal squares have trivial torus character over $\mathbb{F}_3$. The order-three norm gives $D_{bb} = 0$. The inverse-root cocycle identity gives

$$z(x_{ab}(-1)) = 2D + 2bS_{ab} + 2bE_{aa}.$$

Torus conjugation fixes D and negates S_{ab}, whence $D = bE_{aa}$. For distinct $a, c \neq b$, the commuting-root identity for $x_{ab}(1)$ and $x_{cb}(1)$ gives $b_{ab} = b_{cb}$. Thus the coefficient depends only on b. Its values are exactly the coboundary of a diagonal symmetric matrix.

In the exterior-square model, write $z(x_{ab}(1)) = b_{ab}e_a \wedge e_b$. The same commuting-root identity gives $b_{ab} = -b_{cb}$. Three distinct rows outside column b force all coefficients to be zero. Dual tensor squares follow by inverse transpose. This proves (34) on all root generators and hence on G.

D.2 Additive equivariance

An additive map of K-vector spaces is only $\mathbb{F}_p$-linear. After scalar extension from $\mathbb{F}_p$, the underlying additive natural module splits into its Frobenius twists. The same is true for symmetric and exterior squares. In odd characteristic those two squares are irreducible for the special linear group of degree at least five. Distinct-twist products $A_i \otimes A_j$ and $A_i \otimes A_j^*$ are irreducible by Steinberg's tensor-product theorem. Equal-twist endomorphism modules have only scalar and adjoint-heart constituents. These degree-two types differ from the natural type and from the opposite symmetric or exterior-square type.

It follows that an equivariant additive map vanishes unless these types match. In a matching natural, symmetric-square, or exterior-square type it is a scalar multiple of the appropriate entrywise Frobenius map. This rule applies also over $\mathbb{F}_3$; it does not assume that an arbitrary additive map is K-linear. The finite-point degree-two constituent statements are the standard defining-characteristic statements used in the representation-repair appendix; see [8, Section 3.2 and the defining-characteristic proofs of Theorems 2.9 and 2.11].

D.3 Symplectic and split even orthogonal groups

Let $G = \mathrm{Sp}_{2m}(K)$, $m \geq 5$, and write $W = A \oplus A^*$. Its Siegel Levi contains

$$L = \{\mathrm{diag}(g, g^{-T}) : g \in \mathrm{SL}_m(K)\}.$$

Let z have values in $\mathrm{Hom}(W_i, W_j)$, with action $X \mapsto \rho_j(g)X\rho_i(g)^{-1}$. By (34), subtract a coboundary so that $z|_L = 0$. Use

$$u_+(B) = \begin{pmatrix} I & B \\ 0 & I \end{pmatrix}, \qquad u_-(C) = \begin{pmatrix} I & 0 \\ C & I \end{pmatrix}, \qquad B^T = B,\ C^T = C.$$

Write $z(u_+(B))$ in blocks P, Q, R, S. The lower-left block R is additive because upper-unipotent conjugation fixes it. L-equivariance and the preceding type rule force $R = 0$: the symmetric natural-square parameter cannot map to the opposite dual-square target. Then P and S are additive and vanish by their types. Finally Q is additive. It is zero for distinct twists and is

$$Q(B) = \lambda B^{p^i}$$

when $i = j$. A block-scalar coboundary with diagonal entries $\alpha I, \beta I$, $\beta - \alpha = \lambda$, removes it without changing $z|_L$.

Apply the same argument to the lower group. Only the equal-twist value μC^{p^i} can remain in its lower-left block. Choose

$$B = E_{11} - E_{22}, \qquad C = (e_1 + e_2)e_3^T + e_3(e_1 + e_2)^T.$$

Then $BCB = CBC = 0$ and $BC \neq 0$, and direct multiplication gives

$$[u_+(B), u_-(C)] = \mathrm{diag}(I + BC, I - CB) \in L. \tag{35}$$

In the equal-twist dual-number representation, the cocycle fixes L and the upper group, but replaces C by $(1 + \epsilon\mu)C$ in the lower group. Relation (35) therefore forces $\mu BC = 0$, so $\mu = 0$. The Levi and the two full unipotent groups generate the symplectic group. This proves vanishing for every pair of twists, including when a simple exterior-square heart itself has nonzero first cohomology.

For the split even orthogonal group use the same matrices with B,C skew-symmetric. Replace the symmetric-square parameter by the exterior-square parameter in the type argument. For the final commutator take

$$B = E_{12} - E_{21}, \qquad C = E_{23} - E_{32}.$$

Here $BC = E_{13}$, $CB = E_{31}$, and $BCB = CBC = 0$. The elementary orthogonal subgroup so generated has p-prime index in the full orthogonal group. Restriction on first cohomology is injective across that index, by averaging or restriction and corestriction. Thus the same vanishing holds for the full split orthogonal group.

D.4 Even unitary groups

Let K/K_0 be quadratic, write $^-$ for its involution, and let $G = \mathrm{U}_{2m}(K/K_0)$ with $m \geq 5$. Its Siegel Levi is $\mathrm{diag}(g, \bar{g}^{-T})$ and contains $\mathrm{SL}_m(K)$. The upper and lower parameters satisfy $B^\dagger = -B$ and $C^\dagger = -C$. If $|K_0| = p^f$, the additive antihermitian parameter module extends from $\mathbb{F}_p$ to the sum of the f types $A_s \otimes A_{s+f}$; transpose identifies the paired orientations. In the upper-right block of $\mathrm{Hom}(W_i, W_j)$, the matching type occurs only for $i = j$ modulo $2f$. The same block calculation therefore leaves only λB^{p^i} in the equal-twist case, removed by a block-scalar coboundary.

Choose a nonzero $\theta \in K$ with $\bar{\theta} = -\theta$ and use

$$B = \theta(E_{11} - E_{22}), \qquad C = \theta\big((e_1 + e_2)e_3^T + e_3(e_1 + e_2)^T\big).$$

They satisfy the hypotheses of (35). Its diagonal matrix lies in the special-unitary Siegel Levi and kills the remaining lower scalar. The generated elementary special unitary group has p-prime index in the full unitary group, completing the even-dimensional case.

D.5 The anisotropic tail and the short roots

For odd orthogonal, nonsplit even orthogonal, or odd unitary groups, write

$$W = A \oplus A^{\dagger,*} \oplus Z,$$

where $\dim A = m \geq 5$ and Z has anisotropic dimension at most two in the orthogonal case and at most one in the unitary case. Let H denote its form matrix. Let G_0 be the full split-even orthogonal or even-unitary group on the first two summands, times the full tail isometry group. The latter has p-prime order.

The preceding even cases give zero first cohomology on the even whole Hom blocks. A central scalar of p-prime order acts nontrivially on the even natural and dual blocks, so they too have zero first cohomology. The trivial block has zero first cohomology because the large classical abelianization has p-prime order. The direct-product formula therefore permits a coboundary normalization with $z|_{G_0} = 0$.

The upper parabolic radical consists of

$$u(X,B) = \begin{pmatrix} I & B - \frac{1}{2}XH^{-1}X^\dagger & X \\ 0 & I & 0 \\ 0 & -H^{-1}X^\dagger & I \end{pmatrix}, \qquad B^\dagger = -B. \tag{36}$$

Its center $u(0,B)$ lies in G_0 and has zero cocycle. Hence $z(u(X,0))$ is fixed by every central $u(0,B)$. In a three-by-three block matrix, these fixed-space equations force blocks $(2,1),(2,3),(3,1)$ to vanish. The remaining diagonal blocks are unchanged by upper-unipotent conjugation. As the cocycle is zero on the center, they are additive functions of X. Their special-linear types are trivial, adjoint, or distinct-twist tensor types, whereas X is a sum of natural twists. Thus all diagonal blocks vanish.

The remaining off-diagonal blocks $(1,3)$ and $(3,2)$ are additive. Write $X_i = X^{p^i}$ and $H_i = H^{p^i}$. Special-linear equivariance gives

$$R(X) = X_j D, \qquad W(X) = EX_i^\dagger$$

for fixed tail matrices D,E. Tail equivariance says that D intertwines the two twisted tail actions. The commutator of two elements $u(X,0)$ lies in the center. Comparing the two orders of their cocycle products gives

$$X_j(E - DH_i^{-1})Y_i^\dagger = Y_j(E - DH_i^{-1})X_i^\dagger.$$

Take X supported in row one and Y supported in row two. The two sides have different matrix positions, forcing $E = DH_i^{-1}$. The remaining $(1,2)$ block is

$$Q(X) = X_j D H_i^{-1} X_i^\dagger.$$

Indeed the difference is additive, and there is no equivariant additive map from its natural parameter type to that degree-two target type.

These are exactly the blocks of the coboundary of the constant matrix supported on the tail block D. It intertwines G_0, so subtracting it preserves $z|_{G_0} = 0$ and kills the whole upper radical. A Weyl element in G_0 exchanges the isotropic halves, also killing the opposite radical. These radicals and G_0 generate the full group up to a p-prime component quotient. Restriction is injective across that quotient, proving vanishing. Matrix size greater than 12 over the defining field is a conservative bound ensuring that every required isotropic half has dimension at least five.

D.6 Products, the multiplier input, and central quotients

For a general linear factor, restrict to its special linear subgroup; the determinant quotient has p-prime order, so (34) gives the same whole-Hom vanishing. For a product of classical factors, an irreducible bimodule component is either one of the whole small Hom modules above or an external tensor of natural modules for two different factors. In the latter case the direct-product formula for first cohomology gives zero, because both nontrivial natural factors have zero invariant space. Commutative unitary factors have p-prime order. Semisimplicity of the associative enveloping algebra decomposes any genuine bimodule as a direct sum of these modules, with multiplicities and field twists. This proves the first assertion of Lemma D.1.

Here is the precise multiplier input for its second assertion. Kessar–Koshitani–Linckelmann [11, Proposition 7.1], on printed page 10 of the accepted version of *Conjectures of Alperin and Broué for 2-blocks with elementary abelian defect groups of order 8*, lists the exceptional Schur multipliers in odd defining characteristic as

$$A_1(9), \quad {}^2A_3(3), \quad B_3(3), \quad G_2(3).$$

The generic multiplier is the center of the simply connected cover and has p-prime order. All classical exceptions are excluded by the matrix-size cutoff above. An independent primary comparison is Babai–Goodman–Kantor–Luks–Pálfy, *Short Presentations for Finite Groups*, Remark 3.2, printed page 86, and its twisted-group discussion on printed page 95[12].

The natural derived classical group is perfect and differs from its simple quotient by a p-prime central group. The verified multiplier statement gives zero degree-one and degree-two cohomology with trivial characteristic-p coefficients for both groups. The quotient of the full classical group by the natural derived group is p-prime: it is a determinant, norm-one determinant, or determinant/spinor component quotient. Averaging in the Lyndon–Hochschild–Serre sequence preserves both vanishings. Direct products preserve them by the low-degree Künneth formula; additional central p-prime factors have no effect.

Therefore $H^1(G, P) = H^2(G, P) = 0$ for any trivial P. If P is central in a genuine bimodule B, it is trivial under conjugation. The long exact sequence of $0 \to P \to B \to B/P \to 0$ now gives $H^1(G, B/P) = 0$. All coboundary equations can be solved on the supplied generators. This proves Lemma D.1 constructively.

D.7 The scope of whole-chain quotients

For clarity, a stronger quotient observation used in radical-layer applications has a different hypothesis from an arbitrary subquotient. Consider a sum of whole small Hom modules. Separate its semisimple summands and group all isomorphic nonsemisimple chains first. The latter have composition series $1, H_i, 1$, with distinct hearts after this grouping. The actual degree-two image algebra has full independently supported top-to-heart and heart-to-bottom maps for each heart, their top-to-bottom products, and one common scalar character on the trivial coordinates. This is the joint image algebra described in the representation-repair appendix; the independence follows by multiplying its surjection onto each chain by the corresponding heart idempotent on the left or right.

For an invariant submodule K of such a whole sum, its projection to heart i is $H_i \otimes K_i$ for a multiplicity subspace K_i. Applying the independently supported row maps shows that K also contains the corresponding bottom coordinates. Quotient first by these entire radical submodules. The removed copies become trivial top coordinates; the remaining copies are still whole chains. The residual kernel has zero heart projection. The independently supported column maps then force its surviving top coordinates to vanish. It lies entirely in trivial coordinates.

Thus every such quotient is a sum of whole chains and semisimple summands, modulo a trivial subspace. The preceding first- and second-cohomology vanishings apply to it. This argument does not apply to an arbitrary submodule such as an isolated simple heart.

Over $\mathbb{F}_p$, extend scalars to the algebraic closure, introducing the listed Frobenius twists, and group isomorphic whole chains before using heart idempotents. The quotient argument and cohomology vanishing then descend by finite linear equations. In particular, an actual additive subgroup need only be $\mathbb{F}_p$-linear; no unsupported defining-field linearity is assumed.

E Transport with a normal unitary subgroup or a unitary complement

Throughout this appendix, $k = \mathbb{F}_q$ has odd characteristic p. All linear actions are left actions. For an involutive algebra A, write

$$U(A) = \{u \in A^\times : u^*u = 1\}.$$

A genuine A-bimodule has commuting unital left and right associative actions; its unitary action is $x \mapsto uxu^{-1}$. An invariant subquotient certificate consists of such a bimodule V, $U(A)$-invariant subspaces $P \subseteq V_0 \subseteq V$, and an explicit equivariant identification with V_0/P. Neither subspace is required to be an A-bimodule.

We use Theorem C.1, proved in the semisimple appendix, in the following precise form. For semisimple A, an invariant subquotient certificate, and an explicit affine $U(A)$-action with the certified linear part, it returns the transporter from zero to a supplied vector as a union

$$\bigcup_{j=1}^{t} a_j U(B), \qquad B \subseteq A, \qquad t\nu(B/\operatorname{rad} B) \le \nu(A). \tag{37}$$

Here the cosets are distinct, B is one common source star-subalgebra, and all representatives are actual unitary elements, and complete stabilizer generators are returned. The potential satisfies $1 \le \nu(A) \le q^{O(d)}$ if A has faithful degree d. If $B = A$, there is at most one coset and the affine source is fixed by the whole group. The running time is $q^{O(d)}$ times a fixed polynomial in the explicit auxiliary dimensions and $\log q$. In particular, an auxiliary coinduced representation is not a faithful degree parameter. These output and charging properties, rather than only an orbit decision procedure, are needed below.

E.1 Original-degree group computations

A matrix group supplied in faithful degree d over k has a faithful permutation action on the q^d vectors of its natural module. Deterministic polynomial-time permutation-group algorithms therefore give membership, image kernels and preimages for original-degree representations, normal closures, and actual word lifts in $q^{O(d)}$ time. We use the constructive algorithms of Kantor and Luks [10, Section 3]; in particular their normalizing-subgroup centralizers and specified-quotient operations retain original straight-line programs. This use does not enumerate $\mathrm{GL}_d(k)$.

For an additive image, retain a basis over $\mathbb{F}_p$ and actual group words lifting it. An image kernel may equivalently be generated by taking the normal closure of the corrected original generators, commutators of image-basis lifts, and their pth powers. All groups below stay in their original faithful representations.

Finite-dimensional algebra radicals, semisimple factors, star-stable Levi complements, unitary generators, and their constructive evaluations are the standard odd-characteristic finite-field algebra primitives. The deterministic routines of Ivanyos and Qiao, Theorem 7(1) and Section 3[3], and the corresponding star-algebra routines of Brooksbank and Wilson[1] have polynomial dependence on the characteristic, which is absorbed in the displayed q-exponent. Algebra quotients may be stored by structure constants; we never assume that a quotient preserves the minimal faithful degree of its parent.

E.2 An arbitrary normal p-group and a unitary complement

Theorem E.1. *Let F be a semisimple star-algebra with faithful degree N_F. Suppose that a matrix group of faithful degree N_H is supplied as*

$$H = R \rtimes \kappa(U(F)),$$

where R is an actual normal p-subgroup, κ is an injective computable embedding, and the projection and actual lifts are supplied. Let M be an explicit H-module whose restriction to $\kappa(U(F))$ has an invariant subquotient certificate from a genuine F-bimodule. Then the transporter of two vectors of M, including an actual representative and the full source stabilizer, is computable deterministically in

$$q^{O(N_F+N_H)}\,\mathrm{poly}(\textit{input length}, \log q).$$

There is no requirement that R normalize F or act on the ambient bimodule of the certificate.

Proof. Define the R-augmentation filtration

$$M_0 = M, \qquad M_{i+1} = \sum_{r\in R}(r-1)M_i.$$

The supplied generators suffice to compute each sum. Normality of R makes every term H-invariant. The augmentation ideal of a finite p-group in characteristic p is nilpotent, so the filtration strictly decreases to zero and has length at most $\dim M$. The group R acts trivially on each grade.

Fix the source v. With actual target normalizations recorded, maintain the exact prefix

$$H_i = \{h \in H : hv - v \in M_i\}, \quad R_i = H_i \cap R, \quad \pi(H_i) = U(F), \quad w - v \in M_i. \tag{38}$$

No complement in H_i is assumed. The map

$$\tau_i : R_i \longrightarrow M_i/M_{i+1}, \qquad \tau_i(r) = rv - v + M_{i+1}$$

is additive. Compute its entire image B_i and kernel with original word lifts. If $h \in H_i$, conjugating r by h transforms this increment by the grade action of $\pi(h)$: the extra term involves $(r-1)(h^{-1}v - v)$ and belongs to M_{i+1}. Thus B_i is $U(F)$-invariant, although H_i need not split.

For a lift $h_u \in H_i$ of $u \in U(F)$, put

$$Q_i = (M_i/M_{i+1})/B_i, \qquad z_i(u) = h_u v - v + M_{i+1} + B_i.$$

The value is independent of the lift and satisfies the cocycle identity. The grade inherits its subquotient certificate from the original bimodule, so Q_i carries precisely the affine action covered by (37). The desired layer equation is transport from zero to the class of $w - v$.

The additive space B_i may be only $\mathbb{F}_p$-linear. In that case conduct this argument over the prime field. The faithful degrees become $(\log_p q)N_F$ and $(\log_p q)N_H$, and the bound remains $p^{O((N_F+N_H)\log_p q)} = q^{O(N_F+N_H)}$. No permutation-factorial bound in that enlarged degree is used. The external output remains a group of actual original matrices.

Suppose first that the leaf gives a proper $B \subsetneq F$. For each coset representative a_j, find a lift in H_i and correct its translation by an actual R_i-element. We obtain $h_j \in H_i$ with

$$h_j v - w \in M_{i+1}, \qquad \pi(h_j) = a_j.$$

Recurse on the original full vector equation, with source v, target $h_j^{-1}w$, and the actual group

$$H' = R \rtimes \kappa(U(B)). \tag{39}$$

Every true solution in that leaf coset belongs to $h_j H'$. Conversely, every returned full solution in H' gives a true solution after multiplication by h_j. Dropping intermediate prefix restrictions is therefore harmless; final equality imposes them again.

To check closure, let $J_B = \operatorname{rad} B$ and choose an actual star-Levi $L \subseteq B$. The new normal p-kernel and complement are

$$R' = R\kappa(U(1 + J_B)), \qquad \kappa(U(L)).$$

Their intersection is trivial, and $H' = R' \rtimes \kappa(U(L))$. The original certificate restricts to a genuine L-bimodule certificate. All these algebras are actual subalgebras of the original F; no enlarged faithful quotient representation is introduced.

If the leaf has $B = F$, its affine source is fixed by all of $U(F)$. A nonempty target class is consequently zero. Choose $r \in R_i$ with $\tau_i(r) = w - v$ modulo M_{i+1} and replace w by $r^{-1}w$. For each stored generator h_s of H_i, choose $k_s \in R_i$ with

$$\tau_i(k_s) = -(h_s v - v) + M_{i+1}.$$

Then the full next prefix is

$$H_{i+1} = \langle \ker \tau_i,\ k_s h_s \text{ for all } s \rangle. \tag{40}$$

Every displayed generator fixes v modulo M_{i+1}. Their projections generate $U(F)$, and the full projection kernel is contained through $\ker \tau_i$. An arbitrary element of the true next prefix differs from a word in these generators by an element of that kernel. This proves equality, including all commutator relations; corrected generators are not assumed to form a complement.

At the final grade, the prefix is the full source stabilizer. Recorded normalizations give the transporter. The common source subalgebra in (37) gives common recursive source groups in (39). Successful branch representatives and the full branch kernels reconstruct the source stabilizer by taking ratios to one representative.

Each proper call decreases the concrete semisimple-algebra dimension. Each no-loss call advances a filtration of polynomial length. The sum-of-child-potential bound (37) bounds the tree by $q^{O(N_F)}$ times polynomial depth. Every node costs $q^{O(N_F+N_H)}$ times a fixed polynomial. Coinduced auxiliary repairs are local and discarded after original words have been recovered. This proves the bound and the full output assertion. □

Corollary E.2. *Let E be any finite-dimensional star-algebra faithfully represented in degree N. Vector transport in an invariant subquotient of a genuine E-bimodule under $U(E)$, with full stabilizers, is deterministic in $q^{O(N)}$ times a polynomial in the explicit input and* $\log q$.

Proof. Choose a star-Levi F of E and put $J = \operatorname{rad} E$. Then $U(E) = U(1+J) \rtimes U(F)$, and restriction of the original certificate to F meets Theorem E.1. □

E.3 Compatible groups with solvable quotient

Theorem E.3. *Let E be a star-algebra of faithful degree N_E and H a matrix group of faithful degree N_H. Supply an injective computable $\kappa : U(E) \to H$ with normal image and solvable quotient, and star-preserving algebra automorphisms σ_h satisfying*

$$h\kappa(u)h^{-1} = \kappa(\sigma_h(u)), \qquad \sigma_{\kappa(u)}(a) = uau^{-1}.$$

Suppose that V is a genuine E-bimodule with an explicit linear H-action such that

$$h(axb) = \sigma_h(a)h(x)\sigma_h(b), \qquad \kappa(u)x = uxu^{-1}.$$

Let $P \subseteq V_0 \subseteq V$ be H-invariant subspaces. Then vector transport in V_0/P, including actual witnesses and the full source stabilizer, is deterministic in

$$q^{O(N_E+N_H)}\,\mathrm{poly}(\textit{input length}, \log q).$$

Computed unitary elements and all supplied actions must be evaluable with polynomial cost in their explicit dimensions.

Proof. Write $J = \operatorname{rad} E$, choose an actual star-Levi F, and put $R = \kappa(U(1+J))$. This is normal in H. Star-Levi complements are conjugate by unitary elements of $1+J$, constructively. Indeed, solve the linear intertwining equations for $x \in 1+J$ between their identified quotient bases. Separability and Malcev conjugacy give a solution. The element x^*x centralizes the source complement, so $r = x(x^*x)^{-1/2}$ is a unitary conjugator. The inverse square root is computed by an integer power of the unipotent element: if its order divides p^e, choose $2t = -1$ modulo p^e and use $(x^*x)^t$. No characteristic-zero power series is used.

Correct each original generator h by an actual $r_h \in R$ so that $n_h = r_h h$ normalizes F. Together with $C_R(F)$ these elements generate the full normalizer $N = N_H(F)$, and $H = RN$. For if a radical element normalizes F, its trivial quotient action and injectivity of $F \to E/J$ force it to centralize F. Thus $N_R(F) = C_R(F)$, proving the completeness assertion. The group $C_R(F)$ is obtained from the nilpotent star-algebra $J \cap C_E(F)$ by skew-layer Cayley generators. Furthermore

$$N \cap \kappa(U(E)) = \kappa(U(F))C_R(F),$$

with commuting factors of trivial intersection. Hence $N/\kappa(U(F))$ is solvable.

Write $F = \prod_i \mathrm{Mat}_{a_i}(\mathbb{F}_{q^{f_i}})$. The faithful degree gives $\sum_i f_i a_i \le N_E$. First enumerate the actual image of N on its simple factors, then its image on their centers. The factor permutation image is solvable and has order $q^{O(N_E)}$; the center image has order at most $\prod_i f_i = q^{O(N_E)}$. For completeness, a solvable permutation group of degree t, viewed over $\mathbb{F}_p$, has composition-diagonal image of order at most $p^{9t/4}$ by the completely reducible solvable linear-group bound of Halasi–Maróti [9, Theorem 1.2]. Its remaining normal p-subgroup has order at most $p^{v_p(t!)} \le p^{t/(p-1)}$. This proves the required exponential permutation bound without a factorial in N_H.

Every center-fixing remaining automorphism is inner on each matrix factor. Compute implementers by linear equations. A star-preserving implementer a has a^*a central. On a second-kind factor central norms are onto; on a first-kind factor the obstruction is a square class, of order at most two; exchanged factors have no obstruction. The subgroup N_0 whose F-action is unitary-inner therefore has index $q^{O(N_E)}$ in N. Enumerations of these actual images retain lifts and Schreier generators.

For every generator n of N_0, choose $u_n \in U(F)$ implementing its F-action and form $c_n = \kappa(u_n)^{-1}n$. Adjoin $\kappa(U(F) \cap Z(F))$ and let C be the resulting group. Then

$$N_0 = \kappa(U(F))C, \qquad C = \ker(N_0 \to \mathrm{Aut}(F)), \qquad C \cap \kappa(U(F)) = \kappa(U(F) \cap Z(F)).$$

The last group is the algebra-central unitary subgroup, not the abstract center of the finite group $U(F)$. It is abelian, and the remaining quotient embeds in the solvable group $N_0/\kappa(U(F))$. Consequently C is solvable and fixes F pointwise.

Compute a composition series of C's original faithful natural module. The kernel C_p of its complete diagonal action is a normal p-group; the diagonal image has order at most $q^{9N_H/4}$ by the same completely reducible solvable-group bound. This construction is deterministic within $q^{O(N_H)}$: in each quotient, enumerate vectors, take a cyclic span of minimum positive dimension, and repeat. Such a span is simple. Actual diagonal cosets and kernel words are retained.

Consider $H_p = \kappa(U(E))C_p$. Since C_p fixes F pointwise,

$$H_p = (RC_p) \rtimes \kappa(U(F)).$$

Here RC_p is a normal p-group. Its intersection with the displayed complement is trivial: any intersection element is algebra-central in $U(F)$, whose central unitary group has p-prime order, whereas RC_p is a p-group. Restricting the original bimodule certificate to F permits Theorem E.1. This argument requires only invariance of P and V_0, not centrality or pointwise fixation.

There are $q^{O(N_E+N_H)}$ actual cosets of H_p in H. Indeed $H = RN$, $C_R(F) \subseteq N_0$, and the two enumerated indices are $[N : N_0]$ and $[C : C_p]$. Normalize the target by each actual coset representative and invoke the preceding theorem. All successful answers have the same H_p source stabilizer. Its full generators and ratios of successful representatives to one representative generate the full H source stabilizer. The number of calls and their original-degree costs give the asserted bound. □

E.4 Physical subspaces and frame groups

Corollary E.4. *Under Theorem E.1, suppose that H acts on a physical space W whose restriction to $U(F)$ is a supplied genuine F-module. For subspaces $A, A' \subseteq W$ of dimension m, full H-subspace transport is deterministic in*

$$q^{O(N_F+N_H+m)}\,\mathrm{poly}(\textit{input length}, \log q).$$

For several simultaneous frames, replace m by the sum of their dimensions.

Proof. Choose injective frames $X, X' : k^m \to W$ and adjoin an independent $\mathrm{GL}_m(k)$. This is the unitary group of the exchange algebra $B = \mathrm{Mat}_m(k) \times \mathrm{Mat}_m(k)^{\mathrm{op}}$. The frame space $\mathrm{Hom}_k(k^m, W)$ is a genuine $(F \times B)$-bimodule, with F acting on the output and the primal B factor acting on the input. Its unitary action is

$$(h, b)X = hXb^{-1}.$$

The full group has normal p-kernel $R \times 1$ and complement $U(F) \times \mathrm{GL}_m(k)$. The added faithful degrees are O(m). Theorem E.1 therefore gives its full frame transporter. For each h taking A to A', the compatible b is unique because both frames are injective. Projecting witnesses and all stabilizer generators consequently gives the exact subspace transporter. In a physical-cover call only the current row frames are added, and their dimensions sum to O(N); old flags remain in the already computed group rather than being reintroduced as new frames. □

If the original group also has a normal pure kernel $U(E)$, its intersection with the stabilizer of A is exactly $U(E_A)$, where

$$E_A = \{a \in E : aA \subseteq A,\ a^*A \subseteq A\}.$$

This is a computable star-subalgebra. For a family of physical subspaces intersect these conductors. The quotient of the full subspace stabilizer over this pure kernel embeds in the prior quotient, so a p-only state remains p-only.

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